ECDLP cost across whole isogeny classes: prime, binary and Koblitz

curves **Question.** Take one ECDLP instance `Q = kP`, carry it by explicit, verified isogenies to many curves of its isogeny class — spread over the class's endomorphism-ring strata, reached through small and through large ("hard to find") isogeny degrees — and run the same Pollard rho on every curve. Does the curve, its stratum, its level in any ℓ-volcano, the degree or kernel field of the isogeny that reached it, the generator of `⟨P⟩`, or the curve the instance was drawn on change how fast rho solves it? Report min, max, mean and the rest.

Protocol. `PROTOCOL.md`, frozen before the run (sha256 `81c3232bdab481244324101d10e2d50199da1990ceba10c8dc66cc206286409e`); the `controlled-experiment` skill's classes, controls, decision rules and power check are in it. Run: `runs/run-20261018T0857Z/` (command card `COMMANDS.md`, every record and log). Tool: `isoclass-rho` (`src/bin/isoclass_rho/`).

Headline (measured; the per-rule verdicts are in "Reading"). Four classes, 77,900 rho solves on 189 curves plus controls, every solve returning the planted `k`, every isogeny edge verified on points:

- **The curve does not matter to a generic walk.** Across 104 curves of a 347,004-curve prime-field class (12 endomorphism-ring strata, reached through 3-, 5-, 7-, 13-, 19-, 31-, 41-, 43-, 47- and 61-isogenies), 45 curves of a `j = 0` class, 40 curves of a 327,080-curve binary class and the Koblitz class over `F_2^41`, no curve, stratum, volcano level, distance, isogeny degree or kernel-field degree left the preregistered band; the across-curve tests found no heterogeneity; the between-curve spread of means is 1.1–1.2× what the single-solve spread predicts. Isomorphic models and other field bases of the root move a 100-seed mean as much as any isogenous curve does.
- **Neither does the generator or the source curve.** 600 generators of `⟨P⟩` on P1 (and 400, 384 on P0, B1) with the same secret: per-generator means spread exactly as the null predicts (variance ratios 0.84–1.15, KW p ≥ 0.12 per curve), no correlation with `r`, its size or weight or `x(P_i)`; instances drawn on five source curves and carried to every target: η^2 of the source 0.0002–0.0012, of the target 0.008–0.011, neither detectable. Isolated flagged cells (2 of 520, 1 of 384, 1 of 180) are the low tail of many 25-solve means and do not recur for the same curve in any other row.
- **The detector works, and the one real effect is the crater's own.** The planted negation fold gave 0.63–0.77 of the unfolded mean (expected 0.71, $|z|$ 6–10); the planted 4-adding walk 1.27–1.30 ($|z| \approx 4$). On the `j = 0` crater the automorphism-folded walk cost 0.439 [0.401, 0.481] of the plain walk (expected $1/\sqrt{6} = 0.408$, $z = -25$); on the Koblitz crater the Frobenius-folded walk . Those folds exist only on the crater curve; every curve the isogenies lead to has lost them and costs the same as the rest. In wall time the Koblitz fold still wins by 1.7× (τ) and 2.2× (τ and negation), even though each folded step costs 41 squarings more (1.9–2.0 µs per step against 0.50 µs).
- **Hard isogenies are findable and change nothing.** The 409- and 1721-isogenies of the Koblitz curve (division polynomials of degree 83,640 and 1,480,920, out of reach) were found through torsion abscissae in `F_{2^{41·204}}` and `F_{2^{41·215}}` in about 1 min (409) and 5 min (1721) per kernel on a loaded core (three kernels per search: 197 s and 978 s per search); the curves they reach (conductor-409, −1721 and −703,889 strata of a 708,153-curve class) cost 0.978–1.009 of the reference by stratum ($|z| \le 0.8$), and the crater curve itself 0.996. End-to-end, finding the isogeny is the only cost that depends on the route: 7–74 ms for the Elkies edges of P1 against a 98 ms solve, 1.4–3.1 s for the 23- and 31-isogenies of B1 against a 0.19 s solve.

Nothing here is an index-calculus baseline change (CLAUDE.md success criterion): rho against rho, one instrument, and the only speed-ups are the crater curves' own endomorphisms, which an isogeny removes.

Method

`isoclass-rho` (`src/bin/isoclass_rho/`, native Rust; every statistic in this report is computed there, `plot.py` and `make_report.py` only lay out recorded values):

1. **Class arithmetic (exact).** `#E` by baby-step giant-step on random points, checked on eight more; `t`, `D = t² − 4q = f²·D_K`; `h(D_K)` by counting reduced primitive forms; `h(d²D_K)` for every `d | f` by the conductor formula, cross-checked by counting `h(f²D_K)` directly; the class size `Σ_{d|f} h(d²D_K)`.
2. **Walk.** Breadth-first over the active primes from the root, keyed by `j`. A prime is active when it has a rational isogeny: `ℓ | f`, or `ℓ ∤ f` with `(D_K/ℓ) = 1`; inert primes are recorded as inactive (for odd `m` over `F_{2^m}`, every prime of `f` is `≡ ±1 (mod 8)`, which is why the binary and Koblitz classes have no 3- or 5-isogenies). Three ways to find an ℓ-kernel, all ending in the same monic kernel polynomial `h` of degree `(ℓ−1)/2`:
   - *Elkies route* (`F_p`, `j ∉ {0, 1728}`): the roots of `Φ_ℓ(j, Y)` are the neighbours; the library's Elkies construction gives `h` (`crypto_lib::cryptanalysis::isogeny_walk`). `Φ_ℓ` is built natively from the q-expansion of `j` (ℓ = 61: 15 s).
   - *Division route* (any field, any `j`): factor the ℓ-division polynomial over `F_q` into the irreducible factors of degree `d | (ℓ−1)/2`; a root `x(T)` lives in `F_{q^d}`; the x-only ladder gives `x(iT)` there, and `⟨T⟩` is rational exactly when `∏(x − x(iT))` has `F_q` coefficients. The kernel field degree `d` is the "hardness" of the kernel: the smallest extension its abscissae generate.
   - *Torsion route* (large `ℓ | f` off the floor, where the Frobenius is a scalar λ on `E[ℓ]`, so every subgroup is rational): a random abscissa in `F_{q^d}`, `d = ord(±λ)`, is a point of `E` or of its twist over `F_{q^d}`; the Montgomery x-ladder by the matching cofactor gives an ℓ-torsion abscissa, its multiples the kernel, grouped into Frobenius orbits `{x(±iλ^jT)}` whose polynomials are over `F_q`. This finds a 409-isogeny of the Koblitz curve over `F_{2^41}` in `F_{2^{41·204}}` in about a minute, where the 409-division polynomial (degree 83,640) is out of reach; kernels are sampled (three per vertex), not enumerated, and the dual of a sampled edge is found by pushing a second torsion abscissa through it.
   - The map on points is Kohel's normalised formula from `h` over `F_p` (`X = ℓx − 2σ − 2(3x²+a)·h'/h − 4(x³+ax+b)·(h'/h)'`, `Y = y·X'`) and, over `F_{2^m}` with `y² + xy = x³ + ax² + b`, `X = x + xL + x²L²`, `Y = y + x³L₃ + xL + (x + y + x²)(xL² + L)` with `L = h'/h`, `L₃ = L³ + L·D₂h/h + D₃h/h` (Hasse derivatives), codomain `(a, b + σ + σ²)`, `σ = Σ_{Q∈S} x_Q` — derived from Vélu's general formulas and checked on points.
3. **Every edge verified on points**: images on the codomain, `φ(P+Q) = φ(P) + φ(Q)`, `n·φ(P) = O ≠ φ(P)`; `#E` re-counted on every measured curve; every rho solve must return the planted `k` (it did).
4. **Levels and strata.** A curve's level in the ℓ-volcano is found from the number of rational ℓ-kernels (one at the floor) and non-backtracking walks from three neighbours to the floor; its endomorphism-ring conductor relative to the walked primes is `∏ ℓ^{level_ℓ}`, which names its stratum, and the stratum's class number says how many curves share that endomorphism ring.
5. **Instances.** `P` of order `n` and `Q = kP` on the root, pushed down the tree. Shifted instances: `(P_s, kP_s)` drawn on source curves and carried up to the root through dual edges (any rational isogeny back, composed with the model isomorphism; sixth roots at `j = 0`) and down to every curve. Generators: `P_i = r_iP`, `Q_i = r_iQ`, the same `k`.
6. **Rho.** Single walker, 32-adding (Teske), distinguished points `θ = 2⁻⁸`, no negation map, restart after `40·2⁸` DP-free steps (the same walk as `volcano-rho`); trial `s` has the same seed on every arm; all jobs shuffled over 4 threads. Folds (positive controls): the walk steps from the canonical point of an orbit under negation, the order-6 automorphism group of `j = 0`, or the Frobenius τ of a Koblitz curve (with and without negation), with the scalar tracked, look-ahead against 2-cycles and deterministic escapes from longer fruitless cycles.
7. **Controls.** Isomorphic models of the root (`F_p`: `(u⁴a, u⁶b)`; `F_{2^m}`: the same curve in another polynomial basis of the field, since `y ↦ y + sx` leaves every abscissa, and so the walk, unchanged); an A/A replica of the root under an independent seed set; planted effects of known size (negation fold `1/√2`; 4-adding walk `1.137`); the structural folds where the curve admits them.

Results

The four classes

class	field	root curve	#E = h·n	t	D = f²·D_K	h(D_K)	class size	strata walked / measured	walked / measured curves	edges verified
P1	F_p, p = 235212858169	a = 228505936335, b = 82673460332 (j = 3539031742)	9 · 26134722469 (35 bits)	355949	315² · (-8205107), f = 3²·5·7	612	347,004	10 / 12	3137 / 104	168
P0	F_p, p = 217710133177	a = 0, b = 205801627284 (j = 0)	28 · 7775395183 (33 bits)	−931946	27792² · (-3), f = 2⁴·3²·193	1	38,472	9 / 12	68 / 45	44
B1	F_2^37, poly 0x2000000053	a = 89938361934, b = 61869205585 (j = 121701357494)	2 · 68719627817 (37 bits)	−302161	833² · (-660703), f = 7²·17	296	327,080	5 / 6	177 / 40	523
K	F_2^41, poly 0x20000000009	a = 0, b = 1 (j = 1)	4 · 549756390943 (40 bits)	−2308219	703889² · (-7), f = 409·1721	1	708,153	3 / 4	25 / 11	10

Class size is Σ_{d | f} h(d²D_K) with h(D_K) counted from reduced forms and the other class numbers from the conductor formula; h(f²D_K) was also counted directly and agrees in every class (h_full_order_counted = h_full_order_formula in each walk.json).

The isogenies walked, easy and hard

class	ℓ	role	volcano depth, crater	route	kernel field degree d	tree edges realised (neighbour searches)	mean ms per kernel search
P1	3	everywhere	2, split	elkies	1 (eigen-orbit 1)	21 (11510)	0.6
P1	5	everywhere	1, inert	elkies	2 (eigen-orbit 2)	39 (6734)	0.6
P1	7	everywhere	1, inert	elkies	3 (eigen-orbit 3)	47 (6112)	0.8
P1	13	everywhere	0, split	elkies	1, 3 (eigen-orbit 3)	33 (473)	1.5
P1	19	everywhere	0, split	elkies	9 (eigen-orbit 9)	13 (454)	2.4
P1	31	hard (first vertices only)	0, split	elkies	15 (eigen-orbit 15)	3 (5)	199.7
P1	41	hard (first vertices only)	0, split	elkies	4, 5 (eigen-orbit 5)	3 (6)	580.4
P1	43	hard (first vertices only)	0, split	elkies	3, 21 (eigen-orbit 21)	4 (6)	846.5
P1	47	hard (first vertices only)	0, split	elkies	23 (eigen-orbit 23)	3 (5)	1,978.4
P1	61	hard (first vertices only)	0, split	elkies	3, 10 (eigen-orbit 10)	2 (5)	5,544.2
P1	2	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P1	11	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P1	17	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P1	23	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P1	29	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P1	37	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P1	53	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P1	59	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P0	2	everywhere	4, inert	cubic	1 (eigen-orbit 1)	13 (171)	0.4
P0	3	everywhere	2, ramified	division, elkies	1 (eigen-orbit 1)	31 (149)	0.9
P0	7	everywhere	0, split	—	— (eigen-orbit 3)	0 (30)	0.6
P0	13	everywhere	0, split	—	— (eigen-orbit 6)	0 (30)	2.7
P0	5	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
P0	11	inactive	0, inert	inert and ℓ ∤ f: no rational ℓ-isogeny			
B1	7	everywhere	2, inert	division	3 (eigen-orbit 3)	272 (498)	2.6
B1	17	everywhere	1, split	division	8 (eigen-orbit 8)	242 (236)	151.8
B1	23	hard (first vertices only)	0, split	division	11 (eigen-orbit 11)	6 (3)	1,630.7
B1	31	hard (first vertices only)	0, ramified	division	5 (eigen-orbit 5)	3 (4)	2,421.5
K	409	sampled, torsion route	1, inert	torsion	204 (eigen-orbit 204)	5 (4)	197,043.2
K	1721	sampled, torsion route	1, inert	torsion	215 (eigen-orbit 215)	5 (4)	977,759.1

P1: every measured curve, one instance, 100 paired seeds per arm

Reference √(πn/2) = 202,614 (×1.016 for the 32-adding walk: 205,856); 11000 solves, all returned the planted k; rho wall 274.5 s on 4 threads.

arm	kind	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	mean ms	e2e ms	ratio to √(πn/2)	z vs rest
all isogenous	pooled	104	10400	2,082	206,675 ± 2,078	193,529	705,972	0.523	98.2		1.020	

controls	iso / representation	3	300	16,725	221,977 ± 12,929	213,735	580,637	0.515	108.1		1.096	
iso0	iso-control	1	100	16,725	234,700 ± 21,182	224,393	536,413	0.460	113.9	113.9	1.158 [1.059, 1.268]	2.58
iso1	iso-control	1	100	26,042	217,979 ± 22,049	210,376	483,020	0.516	104.3	104.3	1.076 [0.974, 1.185]	1.00
iso2	iso-control	1	100	26,889	213,251 ± 23,873	194,380	580,637	0.571	106.0	106.0	1.052 [0.942, 1.175]	0.54
aa	aa-replica	1	100	28,795	213,450 ± 18,766	205,775	528,357	0.449	102.8	102.8	1.053 [0.967, 1.148]	0.70
plant-neg	plant (expected 0.707)	1	100	4,828	146,841 ± 15,608	134,554	399,006	0.542	77.1	77.1	0.725 [0.651, 0.804]	−7.45
plant-r4	plant (expected 1.137)	1	100	47,689	263,043 ± 27,303	244,394	764,442	0.530	122.3	122.3	1.298 [1.172, 1.440]	4.03

Pooled by endomorphism-ring stratum (conductor of End relative to the walked primes), by level in each volcano, by distance from the root, by the degree and kernel field of the edge that reached the curve:

group	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	ratio to √(πn/2)	z vs rest
conductor:1	8	800	9,641	212,144 ± 7,338	204,054	602,234	0.499	1.047	1.52
conductor:105	8	800	6,616	206,945 ± 7,220	195,771	608,230	0.503	1.021	0.08
conductor:15	8	800	8,281	208,215 ± 7,594	197,695	622,950	0.526	1.028	0.41
conductor:21	8	800	10,202	201,608 ± 7,458	183,510	609,909	0.534	0.995	−1.39
conductor:3	16	1600	6,526	206,365 ± 5,391	190,088	608,792	0.533	1.019	−0.12
conductor:315	8	800	5,650	209,999 ± 7,352	202,514	603,463	0.505	1.036	0.92
conductor:35	8	800	2,372	201,269 ± 7,395	184,966	552,691	0.530	0.993	−1.49
conductor:45	8	800	7,889	200,839 ± 7,461	186,744	678,581	0.536	0.991	−1.60
conductor:5	8	800	5,526	206,804 ± 7,578	197,208	678,260	0.529	1.021	0.03
conductor:63	8	800	6,276	204,788 ± 7,453	190,399	676,436	0.525	1.011	−0.52
conductor:7	8	800	10,945	208,763 ± 7,647	198,914	652,271	0.529	1.030	0.56
conductor:9	8	800	2,082	212,672 ± 7,651	197,922	705,972	0.519	1.050	1.60
dist:0	1	100	18,040	197,851 ± 22,960	173,365	471,750	0.592	0.976	−0.76
dist:1	12	1200	2,082	202,692 ± 5,967	190,088	608,792	0.520	1.000	−1.39
dist:2	11	1100	7,889	209,234 ± 6,361	191,028	577,757	0.514	1.033	0.83
dist:3	80	8000	2,372	207,031 ± 2,377	194,231	705,972	0.524	1.022	0.62
ell3:level0	32	3200	2,372	207,245 ± 3,746	194,779	678,260	0.522	1.023	0.36
ell3:level1	40	4000	6,526	205,900 ± 3,356	191,152	622,950	0.526	1.016	−0.58
ell3:level2	32	3200	2,082	207,074 ± 3,742	194,686	705,972	0.521	1.022	0.25
ell5:level0	56	5600	2,082	207,529 ± 2,851	193,233	705,972	0.525	1.024	0.87
ell5:level1	48	4800	2,372	205,679 ± 3,035	193,925	678,581	0.522	1.015	−0.87
ell7:level0	56	5600	2,082	207,629 ± 2,855	194,654	705,972	0.525	1.025	0.97
ell7:level1	48	4800	2,372	205,562 ± 3,030	192,427	676,436	0.521	1.015	−0.97
field_degree:1	17	1700	2,082	208,470 ± 5,180	193,233	622,950	0.523	1.029	0.74
field_degree:15	2	200	23,943	194,685 ± 14,733	179,677	544,465	0.546	0.961	−1.61
field_degree:2	14	1400	5,526	205,051 ± 5,653	194,052	678,581	0.526	1.012	−0.61
field_degree:21	1	100	8,420	200,065 ± 20,366	192,051	488,295	0.519	0.987	−0.64
field_degree:23	2	200	15,547	214,330 ± 15,717	199,012	608,792	0.529	1.058	0.96
field_degree:3	51	5100	2,372	206,958 ± 2,965	194,231	705,972	0.522	1.021	0.26
field_degree:4	1	100	9,319	192,450 ± 21,976	165,708	529,751	0.583	0.950	−1.28
field_degree:5	2	200	19,076	210,860 ± 15,039	198,351	491,788	0.515	1.041	0.55
field_degree:9	13	1300	8,458	207,274 ± 5,784	194,069	608,230	0.513	1.023	0.22
parent_ell:13	23	2300	4,542	207,290 ± 4,576	191,875	705,972	0.540	1.023	0.30
parent_ell:19	13	1300	8,458	207,274 ± 5,784	194,069	608,230	0.513	1.023	0.22
parent_ell:3	8	800	2,082	211,717 ± 7,430	203,153	602,234	0.506	1.045	1.38
parent_ell:31	2	200	23,943	194,685 ± 14,733	179,677	544,465	0.546	0.961	−1.61
parent_ell:41	3	300	9,319	204,723 ± 12,435	190,654	529,751	0.537	1.010	−0.31
parent_ell:43	2	200	8,420	203,855 ± 13,157	191,758	488,295	0.466	1.006	−0.42
parent_ell:47	2	200	15,547	214,330 ± 15,717	199,012	608,792	0.529	1.058	0.96

parent_ell:5	14	1400	5,526	205,051 ± 5,653	194,052	678,581	0.526	1.012	−0.61
parent_ell:7	36	3600	2,372	206,382 ± 3,484	194,164	652,271	0.517	1.019	−0.20

Across the 104 measured curves: Kruskal–Wallis H = 122.83 (df 103, p = 0.089); ANOVA F = 1.183 (p = 0.101); between-curve variance ratio 1.183 (observed sd of curve means 11,751 against 10,814 expected from the pooled single-solve spread); largest |z| 3.03 against the Bonferroni threshold 3.49; curves beyond it: none.

Per curve (iterations = group operations; `d` = kernel field degree of the edge that reached it; transport = ms to find and apply the isogenies from the root):

curve	stratum f	levels	path (ℓ)	d	transport ms	min	mean ± CI	median	max	p10	p90	sd/mean	skew	ns/op	e2e ms	ratio	z
v3	1	3:0,5:0,7:0	3	1	8.0	13,298	194,668 ± 20,813	180,774	501,181	77,335	352,530	0.545	0.65	503.6	104.3	0.961	−1.14
v320	1	3:0,5:0,7:0	43·3	1	14.8	21,047	208,491 ± 19,276	190,404	472,963	83,991	354,974	0.472	0.48	468.1	110.5	1.029	0.19
v1083	1	3:0,5:0,7:0	3·3·13	1	10.5	41,824	208,238 ± 19,408	208,331	422,515	75,159	346,441	0.476	0.19	472.4	108.5	1.028	0.16
v1646	1	3:0,5:0,7:0	31·3·3	1	8.8	25,452	217,961 ± 22,158	209,012	529,253	78,718	374,543	0.519	0.54	476.3	113.0	1.076	1.00
v1948	1	3:0,5:0,7:0	41·3·3	1	14.4	9,849	219,835 ± 23,149	209,335	602,234	78,889	386,209	0.537	0.69	487.2	122.3	1.085	1.12
v2231	1	3:0,5:0,7:0	43·3·3	1	15.5	9,641	222,690 ± 21,601	218,673	493,838	94,402	375,125	0.495	0.43	463.7	116.7	1.099	1.46
v2534	1	3:0,5:0,7:0	47·3·3	1	19.6	15,105	202,241 ± 19,611	196,311	508,319	78,807	325,308	0.495	0.40	508.7	120.2	0.998	−0.44
v2835	1	3:0,5:0,7:0	61·3·3	1	34.9	28,934	223,026 ± 19,699	215,777	524,208	100,289	342,584	0.451	0.33	465.2	136.9	1.101	1.63
v0	3	3:1,5:0,7:0	root	—	0.0	18,040	197,851 ± 22,960	173,365	471,750	58,524	361,612	0.592	0.59	429.3	85.8	0.976	−0.76
v23	3	3:1,5:0,7:0	31	15	6.8	23,943	211,321 ± 22,185	205,155	544,465	60,878	354,159	0.536	0.63	462.1	105.8	1.043	0.41
v24	3	3:1,5:0,7:0	31	15	12.5	26,675	178,050 ± 18,947	156,010	476,729	78,962	325,564	0.543	0.75	463.6	95.0	0.879	−2.97
v25	3	3:1,5:0,7:0	41	5	12.5	19,076	206,684 ± 21,114	198,217	464,068	75,803	338,215	0.521	0.43	484.4	112.3	1.020	0.00
v26	3	3:1,5:0,7:0	41	4	21.8	9,319	192,450 ± 21,976	165,708	529,751	63,377	341,471	0.583	0.71	454.6	110.5	0.950	−1.28
v27	3	3:1,5:0,7:0	43	3	14.0	47,465	207,645 ± 16,734	190,161	418,579	110,543	325,058	0.411	0.51	486.2	113.1	1.025	0.11
v28	3	3:1,5:0,7:0	43	21	24.1	8,420	200,065 ± 20,366	192,051	488,295	71,326	347,934	0.519	0.54	468.3	119.2	0.987	−0.64
v29	3	3:1,5:0,7:0	47	23	17.9	15,547	220,490 ± 23,498	208,986	608,792	84,433	383,936	0.544	0.70	484.6	124.4	1.088	1.16
v30	3	3:1,5:0,7:0	47	23	35.2	27,941	208,171 ± 20,928	199,012	530,983	72,500	349,466	0.513	0.50	461.2	130.6	1.027	0.14
v237	3	3:1,5:0,7:0	13·19	9	9.3	18,550	208,474 ± 22,715	189,577	550,531	80,486	375,946	0.556	0.77	483.1	109.9	1.029	0.16
v339	3	3:1,5:0,7:0	43·19	9	19.6	35,772	223,457 ± 22,692	205,300	516,491	96,691	393,632	0.518	0.60	470.7	122.7	1.103	1.46
v1042	3	3:1,5:0,7:0	3·3·13	3	11.2	6,526	214,105 ± 23,566	193,730	542,455	70,496	373,477	0.562	0.50	488.1	114.5	1.057	0.62
v1772	3	3:1,5:0,7:0	31·13·19	9	12.5	14,624	193,884 ± 20,399	173,623	501,154	77,365	341,789	0.537	0.62	473.6	100.3	0.957	−1.23
v2077	3	3:1,5:0,7:0	41·13·19	9	18.8	12,475	209,758 ± 21,847	202,990	488,827	76,952	355,752	0.531	0.37	505.6	122.7	1.035	0.28
v2525	3	3:1,5:0,7:0	43·13·13	3	27.3	21,546	217,882 ± 22,754	189,003	524,402	83,926	373,976	0.533	0.55	471.6	130.1	1.075	0.97
v2831	3	3:1,5:0,7:0	47·13·19	9	40.0	8,458	211,558 ± 21,051	208,124	464,648	73,770	349,057	0.508	0.24	461.2	137.6	1.044	0.46
v92	5	3:0,5:1,7:0	3·5	2	9.7	16,798	218,335 ± 19,162	207,181	451,220	99,181	353,069	0.448	0.24	504.8	119.6	1.078	1.20
v1069	5	3:0,5:1,7:0	3·3·5	2	10.4	19,090	217,958 ± 19,019	213,290	481,363	92,255	353,143	0.445	0.12	510.1	120.0	1.076	1.17
v1110	5	3:0,5:1,7:0	3·5·19	9	14.3	31,485	205,526 ± 21,327	191,743	515,553	82,939	348,206	0.529	0.69	484.6	109.7	1.014	−0.11
v1155	5	3:0,5:1,7:0	3·5·13	3	12.1	9,034	205,324 ± 24,872	185,274	678,260	65,833	373,013	0.618	1.15	494.2	111.1	1.013	−0.11
v1802	5	3:0,5:1,7:0	31·3·5	2	15.4	14,667	212,324 ± 23,720	197,253	574,780	69,879	378,355	0.570	0.64	466.5	114.3	1.048	0.47
v2086	5	3:0,5:1,7:0	41·3·5	2	24.7	5,526	200,960 ± 22,338	185,760	525,884	54,708	354,070	0.567	0.63	476.9	116.8	0.992	−0.50
v2389	5	3:0,5:1,7:0	43·3·5	2	27.5	12,251	194,882 ± 21,671	184,431	536,971	55,025	342,407	0.567	0.44	472.8	123.2	0.962	−1.07
v2710	5	3:0,5:1,7:0	47·3·5	2	38.9	11,101	199,122 ± 18,807	202,191	408,363	75,168	325,952	0.482	0.28	464.3	129.9	0.983	−0.79
v98	7	3:0,5:0,7:1	3·7	3	10.4	38,574	223,395 ± 22,466	204,717	577,757	97,264	386,324	0.513	0.70	486.8	117.1	1.103	1.47
v1076	7	3:0,5:0,7:1	3·3·7	3	12.4	20,428	207,768 ± 22,064	208,857	503,767	57,550	348,381	0.542	0.42	486.5	110.6	1.025	0.10
v1169	7	3:0,5:0,7:1	3·7·19	9	13.6	20,378	206,205 ± 20,456	200,849	482,618	78,693	353,980	0.506	0.40	491.9	114.0	1.018	−0.05
v1186	7	3:0,5:0,7:1	3·7·19	9	16.9	23,445	190,115 ± 18,637	179,775	531,945	70,969	324,332	0.500	0.80	486.3	109.6	0.938	−1.75
v1809	7	3:0,5:0,7:1	31·3·7	3	16.4	10,945	221,310 ± 23,137	219,641	595,983	88,295	348,937	0.533	0.57	491.3	127.0	1.092	1.25
v2094	7	3:0,5:0,7:1	41·3·7	3	26.0	11,449	193,048 ± 19,820	182,724	497,289	68,931	345,965	0.524	0.56	462.9	115.1	0.953	−1.35
v2397	7	3:0,5:0,7:1	43·3·7	3	28.5	23,048	204,250 ± 21,854	195,121	652,271	63,585	326,413	0.546	0.85	489.0	128.5	1.008	−0.22
v2718	7	3:0,5:0,7:1	47·3·7	3	40.2	21,102	224,014 ± 23,762	210,176	576,287	82,126	386,657	0.541	0.65	477.8	145.4	1.106	1.44
v1	9	3:2,5:0,7:0	3	1	7.9	2,082	204,826 ± 21,518	195,304	501,418	77,349	347,650	0.536	0.35	501.8	106.7	1.011	−0.17
v82	9	3:2,5:0,7:0	3·41	5	34.4	53,547	215,036 ± 21,498	199,265	491,788	87,553	366,848	0.510	0.67	479.7	138.8	1.061	0.77

v547	9	3:2,5:0,7:0	3·13·13	1	12.1	36,571	216,391 ± 22,277	190,045	567,769	93,726	384,229	0.525	0.85	467.5	115.6	1.068	0.86
v641	9	3:2,5:0,7:0	3·43·19	9	27.5	32,306	209,599 ± 20,663	205,122	533,891	87,609	358,270	0.503	0.66	472.4	126.9	1.034	0.28
v839	9	3:2,5:0,7:0	3·13·13	3	12.1	4,542	224,837 ± 23,499	217,529	705,972	86,880	374,553	0.533	0.84	488.9	121.3	1.110	1.52
v933	9	3:2,5:0,7:0	3·43·19	9	28.1	15,241	216,069 ± 21,692	206,068	571,037	85,834	365,663	0.512	0.59	487.7	132.5	1.066	0.85
v1303	9	3:2,5:0,7:0	3·13·13	1	11.0	28,917	219,330 ± 20,151	207,346	554,078	91,634	347,848	0.469	0.55	481.0	114.3	1.083	1.24
v2266	9	3:2,5:0,7:0	43·3·13	1	16.3	18,603	195,286 ± 21,897	176,834	586,152	75,866	347,023	0.572	0.93	467.1	107.0	0.964	−1.02
v5	15	3:1,5:1,7:0	5	2	1.5	15,898	205,012 ± 19,775	207,190	505,206	67,259	332,647	0.492	0.31	477.0	98.6	1.012	−0.17
v302	15	3:1,5:1,7:0	41·5	2	23.5	20,703	220,347 ± 23,554	205,110	543,350	82,531	392,668	0.545	0.62	473.3	128.6	1.088	1.14
v1343	15	3:1,5:1,7:0	5·13·13	3	5.2	8,281	209,825 ± 22,030	203,807	493,306	79,331	385,084	0.536	0.49	458.9	101.2	1.036	0.28
v1674	15	3:1,5:1,7:0	31·5·13	3	9.8	45,852	215,037 ± 22,942	200,792	602,622	81,059	381,543	0.544	0.82	455.6	107.9	1.061	0.72
v1976	15	3:1,5:1,7:0	41·5·13	3	15.5	12,214	197,221 ± 20,684	182,108	407,960	71,771	366,998	0.535	0.42	483.8	107.6	0.973	−0.90
v2278	15	3:1,5:1,7:0	43·5·13	3	17.1	25,532	214,635 ± 18,591	217,744	459,334	96,828	337,138	0.442	0.32	469.3	118.0	1.059	0.84
v2580	15	3:1,5:1,7:0	47·5·13	1	21.3	21,119	192,274 ± 23,109	172,306	622,950	68,237	358,547	0.613	1.07	489.4	110.5	0.949	−1.23
v2882	15	3:1,5:1,7:0	61·5·13	3	36.2	21,339	211,369 ± 20,831	206,949	499,016	77,358	348,049	0.503	0.49	481.2	139.1	1.043	0.44
v11	21	3:1,5:0,7:1	7	3	2.5	46,970	202,926 ± 19,347	187,090	493,322	86,670	330,592	0.486	0.64	482.9	102.6	1.002	−0.38
v308	21	3:1,5:0,7:1	41·7	3	24.6	17,268	190,765 ± 19,661	165,413	552,649	93,232	321,870	0.526	1.05	461.0	111.8	0.942	−1.59
v1551	21	3:1,5:0,7:1	7·13·13	1	6.6	29,105	190,314 ± 21,058	177,829	584,676	73,124	332,153	0.565	1.11	497.6	102.8	0.939	−1.53
v1738	21	3:1,5:0,7:1	31·7·13	1	10.2	27,719	196,110 ± 21,195	174,414	558,932	80,424	342,276	0.551	0.96	486.1	106.5	0.968	−0.98
v2040	21	3:1,5:0,7:1	41·7·13	3	16.4	10,202	193,775 ± 22,300	176,222	609,909	66,693	346,741	0.587	0.98	489.2	109.2	0.956	−1.14
v2342	21	3:1,5:0,7:1	43·7·13	1	18.1	37,636	195,581 ± 20,026	178,155	511,715	78,941	330,894	0.522	0.75	463.7	109.4	0.965	−1.09
v2644	21	3:1,5:0,7:1	47·7·13	3	22.2	13,308	206,674 ± 19,554	196,573	468,886	80,054	328,711	0.483	0.27	488.9	119.1	1.020	−0.00
v2946	21	3:1,5:0,7:1	61·7·13	1	36.3	26,634	236,720 ± 24,359	219,306	562,306	95,811	390,916	0.525	0.87	447.7	143.2	1.168	2.43
v1087	35	3:0,5:1,7:1	3·5·7	3	13.3	20,668	202,867 ± 20,181	187,083	518,960	79,865	344,527	0.508	0.47	487.6	112.0	1.001	−0.37
v1093	35	3:0,5:1,7:1	3·5·7	3	14.9	21,101	206,259 ± 19,612	214,510	499,934	85,424	330,228	0.485	0.38	502.3	113.8	1.018	−0.04
v1103	35	3:0,5:1,7:1	3·5·7	3	12.8	21,050	206,461 ± 21,199	192,353	518,953	68,500	336,549	0.524	0.36	501.3	113.6	1.019	−0.02
v1113	35	3:0,5:1,7:1	3·5·7	3	12.4	34,710	213,456 ± 21,336	188,338	552,691	80,315	354,778	0.510	0.45	479.5	114.1	1.054	0.63
v1123	35	3:0,5:1,7:1	3·5·7	3	12.1	34,229	199,452 ± 20,510	189,202	488,319	79,062	333,423	0.525	0.63	479.6	104.9	0.984	−0.69
v1129	35	3:0,5:1,7:1	3·5·7	3	13.5	11,552	201,406 ± 22,369	175,493	520,070	60,850	356,070	0.567	0.54	473.4	108.1	0.994	−0.46
v1139	35	3:0,5:1,7:1	3·5·7	3	14.4	12,965	204,584 ± 21,730	179,063	502,034	76,078	357,267	0.542	0.76	472.8	109.6	1.010	−0.19
v1149	35	3:0,5:1,7:1	3·5·7	3	13.2	2,372	175,665 ± 20,168	162,274	443,181	61,340	327,654	0.586	0.72	467.7	95.6	0.867	−3.03
v33	45	3:2,5:1,7:0	3·5	2	9.5	7,889	187,385 ± 19,672	169,661	409,033	63,116	332,108	0.536	0.37	476.6	97.5	0.925	−1.93
v487	45	3:2,5:1,7:0	3·5·19	9	22.9	12,535	205,533 ± 19,073	206,262	432,284	76,612	324,357	0.473	0.21	448.5	117.6	1.014	−0.12
v625	45	3:2,5:1,7:0	3·43·5	2	24.6	34,137	205,346 ± 21,523	179,958	616,005	79,145	351,940	0.535	0.92	451.2	116.4	1.013	−0.12
v719	45	3:2,5:1,7:0	3·61·5	2	71.8	33,780	210,074 ± 20,564	192,608	499,031	82,926	359,385	0.499	0.66	459.9	166.6	1.037	0.33
v865	45	3:2,5:1,7:0	3·31·5	2	23.2	25,200	196,686 ± 22,287	182,373	612,762	66,550	348,168	0.578	1.01	461.1	114.4	0.971	−0.88
v970	45	3:2,5:1,7:0	3·47·5	2	45.6	10,962	189,863 ± 18,735	179,749	536,963	75,260	313,911	0.503	0.67	484.7	136.5	0.937	−1.77
v1255	45	3:2,5:1,7:0	3·5·13	3	11.4	23,365	199,413 ± 22,396	174,523	540,092	67,995	357,147	0.573	0.64	482.1	109.5	0.984	−0.64
v2254	45	3:2,5:1,7:0	43·3·5	2	16.6	15,281	212,413 ± 24,189	200,916	678,581	72,070	368,981	0.581	1.22	522.6	126.6	1.048	0.47
v39	63	3:2,5:0,7:1	3·7	3	9.9	10,253	205,803 ± 21,754	180,445	504,640	83,742	382,060	0.539	0.81	502.3	113.7	1.016	−0.08
v533	63	3:2,5:0,7:1	3·7·13	3	14.4	6,276	196,257 ± 24,593	186,774	676,436	57,672	334,203	0.639	1.13	485.1	107.6	0.969	−0.84
v632	63	3:2,5:0,7:1	3·43·7	3	26.5	17,525	186,976 ± 19,756	180,849	555,088	64,834	304,626	0.539	0.58	486.8	114.8	0.923	−1.96
v727	63	3:2,5:0,7:1	3·61·7	3	73.9	33,036	200,557 ± 20,420	187,919	463,424	81,785	339,785	0.519	0.68	485.1	170.1	0.990	−0.59
v872	63	3:2,5:0,7:1	3·31·7	3	24.4	14,336	239,823 ± 22,656	233,844	637,573	105,623	370,904	0.482	0.73	482.1	137.8	1.184	2.88
v977	63	3:2,5:0,7:1	3·47·7	3	46.3	11,777	190,879 ± 17,386	194,508	425,647	65,483	292,885	0.465	0.05	490.0	138.3	0.942	−1.79
v1293	63	3:2,5:0,7:1	3·7·19	9	14.5	30,615	199,456 ± 18,599	185,877	440,786	94,593	327,599	0.476	0.56	472.7	106.5	0.984	−0.76
v2261	63	3:2,5:0,7:1	43·3·7	3	18.2	24,658	218,556 ± 21,451	203,796	540,884	93,352	356,754	0.501	0.90	483.1	122.8	1.079	1.09
v128	105	3:1,5:1,7:1	5·7	3	3.5	13,182	200,085 ± 18,507	198,231	517,006	79,406	320,699	0.472	0.52	461.9	95.8	0.988	−0.70
v1361	105	3:1,5:1,7:1	5·7·19	9	6.7	18,814	214,925 ± 21,772	197,494	608,230	81,684	368,878	0.517	0.72	489.9	111.7	1.061	0.75
v1475	105	3:1,5:1,7:1	5·7·13	3	5.9	30,728	211,079 ± 20,846	206,450	487,975	77,432	356,188	0.504	0.58	456.3	101.2	1.042	0.42

v1708	105	3:1,5:1,7:1	31·5·7	3	12.1	15,974	213,976 ± 20,292	204,566	570,495	100,065	336,146	0.484	0.87	489.6	115.7	1.056	0.71
v2004	105	3:1,5:1,7:1	41·5·7	3	16.3	26,556	225,152 ± 22,413	204,186	545,747	85,676	385,472	0.508	0.60	473.9	120.9	1.111	1.62
v2296	105	3:1,5:1,7:1	43·5·7	3	18.3	10,213	186,436 ± 18,811	177,819	424,095	63,146	316,199	0.515	0.43	482.0	108.2	0.920	−2.12
v2588	105	3:1,5:1,7:1	47·5·7	3	23.4	13,871	193,983 ± 19,148	183,714	422,605	82,748	342,639	0.504	0.36	470.2	115.0	0.957	−1.30
v2880	105	3:1,5:1,7:1	61·5·7	3	38.4	6,616	209,922 ± 20,912	209,591	470,241	74,878	356,691	0.508	0.31	486.6	139.0	1.036	0.31
v440	315	3:2,5:1,7:1	3·5·7	3	12.2	39,694	213,567 ± 22,457	196,114	522,575	80,440	376,251	0.536	0.61	477.6	116.0	1.054	0.60
v466	315	3:2,5:1,7:1	3·5·7	3	12.6	12,033	214,134 ± 19,576	202,376	466,224	87,958	344,710	0.466	0.38	476.3	114.2	1.057	0.75
v492	315	3:2,5:1,7:1	3·5·7	3	15.3	17,860	189,253 ± 20,092	185,824	515,533	75,663	307,162	0.542	0.84	497.4	108.6	0.934	−1.71
v738	315	3:2,5:1,7:1	3·5·7	3	12.9	5,650	209,262 ± 20,826	206,145	603,463	59,626	343,578	0.508	0.43	466.0	108.7	1.033	0.24
v768	315	3:2,5:1,7:1	3·5·7	3	11.8	40,644	224,284 ± 22,543	222,530	545,071	78,374	379,066	0.513	0.49	483.6	119.1	1.107	1.54
v794	315	3:2,5:1,7:1	3·5·7	3	12.6	41,514	214,728 ± 20,013	215,945	474,555	83,960	325,550	0.476	0.33	508.5	121.0	1.060	0.79
v1215	315	3:2,5:1,7:1	3·5·7	3	13.1	8,274	214,767 ± 20,482	219,906	424,972	82,961	373,593	0.487	0.10	501.0	117.7	1.060	0.78
v1241	315	3:2,5:1,7:1	3·5·7	3	13.3	21,433	199,994 ± 20,131	202,858	472,628	65,644	324,718	0.514	0.44	482.9	108.3	0.987	−0.65

P0: every measured curve, one instance, 100 paired seeds per arm

Reference √(πn/2) = 110,515 (×1.016 for the 32-adding walk: 112,283); 5200 solves, all returned the planted k; rho wall 68.3 s on 4 threads.

arm	kind	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	mean ms	e2e ms	ratio to √(πn/2)	z vs rest
all isogenous	pooled	45	4500	1,645	110,942 ± 1,720	103,867	390,832	0.531	53.0		1.004	
controls	iso / representation	3	300	3,714	115,805 ± 6,578	107,440	274,089	0.502	55.7		1.048	
iso0	iso-control	1	100	3,714	117,877 ± 11,502	110,677	263,896	0.498	56.5	56.5	1.067 [0.969, 1.173]	1.17
iso1	iso-control	1	100	11,500	116,476 ± 12,363	108,712	252,756	0.542	57.6	57.6	1.054 [0.950, 1.166]	0.87
iso2	iso-control	1	100	26,809	113,060 ± 10,319	102,877	274,089	0.466	52.9	52.9	1.023 [0.934, 1.122]	0.40
aa	aa-replica	1	100	13,639	106,523 ± 10,287	101,260	235,094	0.493	51.6	51.6	0.964 [0.878, 1.059]	−0.83
plant-neg	plant (expected 0.707)	1	100	3,558	70,056 ± 7,845	63,548	212,452	0.571	35.8	35.8	0.634 [0.567, 0.709]	−9.98
plant-r4	plant (expected 1.137)	1	100	11,882	139,777 ± 15,320	138,046	376,412	0.559	61.5	61.5	1.265 [1.136, 1.411]	3.67
fold-aut	fold (expected 0.408)	1	100	5,490	48,534 ± 4,524	46,033	120,502	0.476	26.2	26.2	0.439 [0.401, 0.481]	−25.27

Pooled by endomorphism-ring stratum (conductor of End relative to the walked primes), by level in each volcano, by distance from the root, by the degree and kernel field of the edge that reached the curve:

group	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	ratio to √(πn/2)	z vs rest
conductor:1	1	100	14,535	107,356 ± 11,848	103,482	261,129	0.563	0.971	−0.60
conductor:12	6	600	4,396	109,898 ± 4,594	103,138	330,338	0.522	0.994	−0.48
conductor:16	6	600	2,640	110,445 ± 4,626	104,755	315,041	0.524	0.999	−0.23
conductor:18	6	600	2,966	110,252 ± 4,752	100,474	307,333	0.539	0.998	−0.31
conductor:2	1	100	14,335	111,242 ± 11,062	105,007	278,778	0.507	1.007	0.05
conductor:24	6	600	6,862	111,154 ± 4,749	103,325	333,980	0.534	1.006	0.09
conductor:3	1	100	15,231	116,467 ± 11,843	114,854	319,034	0.519	1.054	0.93
conductor:36	6	600	5,569	113,707 ± 4,855	107,352	390,832	0.534	1.029	1.20
conductor:4	2	200	4,318	104,081 ± 8,229	101,636	300,294	0.570	0.942	−1.67
conductor:6	3	300	6,702	109,997 ± 6,925	102,762	303,373	0.556	0.995	−0.28
conductor:8	4	400	1,645	110,903 ± 5,688	102,764	275,017	0.523	1.004	−0.01
conductor:9	3	300	8,253	114,267 ± 6,401	104,247	296,521	0.495	1.034	1.05
dist:0	1	100	14,535	107,356 ± 11,848	103,482	261,129	0.563	0.971	−0.60
dist:1	2	200	14,335	113,855 ± 8,091	108,893	319,034	0.513	1.030	0.72
dist:2	8	800	4,318	110,119 ± 4,095	102,804	303,373	0.537	0.996	−0.43
dist:3	16	1600	1,645	110,282 ± 2,856	102,504	330,338	0.528	0.998	−0.56
dist:4	18	1800	2,640	111,769 ± 2,738	105,722	390,832	0.530	1.011	0.77
ell2:level0	5	500	8,253	113,325 ± 5,093	107,079	319,034	0.513	1.025	0.97
ell2:level1	10	1000	2,966	110,274 ± 3,694	102,361	307,333	0.540	0.998	−0.40
ell2:level2	14	1400	4,318	110,700 ± 3,099	105,396	390,832	0.534	1.002	−0.18

ell2:level3	10	1000	1,645	111,054 ± 3,645	102,979	333,980	0.530	1.005	0.07
ell2:level4	6	600	2,640	110,445 ± 4,626	104,755	315,041	0.524	0.999	−0.23
ell3:level0	14	1400	1,645	109,503 ± 3,047	103,331	315,041	0.531	0.991	−1.11
ell3:level1	16	1600	4,396	110,798 ± 2,892	104,368	333,980	0.533	1.003	−0.12
ell3:level2	15	1500	2,966	112,437 ± 3,003	104,241	390,832	0.528	1.017	1.20
field_degree:1	44	4400	1,645	111,023 ± 1,738	103,868	390,832	0.530	1.005	0.60
parent_ell:2	13	1300	1,645	109,668 ± 3,153	103,331	315,041	0.529	0.992	−0.93
parent_ell:3	31	3100	2,966	111,591 ± 2,083	104,247	390,832	0.530	1.010	1.11

Across the 45 measured curves: Kruskal–Wallis H = 53.17 (df 44, p = 0.162); ANOVA F = 1.125 (p = 0.264); between-curve variance ratio 1.125 (observed sd of curve means 6,240 against 5,886 expected from the pooled single-solve spread); largest |z| 2.42 against the Bonferroni threshold 3.26; curves beyond it: none.

Per curve (iterations = group operations; d = kernel field degree of the edge that reached it; transport = ms to find and apply the isogenies from the root):

curve	stratum f	levels	path (f)	d	transport ms	min	mean ± CI	median	max	p10	p90	sd/mean	skew	ns/op	e2e ms	ratio	z
v0	1	2:0,3:0	root	—	0.0	14,535	107,356 ± 11,848	103,482	261,129	36,694	198,046	0.563	0.62	499.8	52.5	0.971	−0.60
v1	2	2:1,3:0	2	1	0.1	14,335	111,242 ± 11,062	105,007	278,778	43,155	182,680	0.507	0.92	470.7	51.9	1.007	0.05
v2	3	2:0,3:1	3	1	0.1	15,231	116,467 ± 11,843	114,854	319,034	40,894	177,294	0.519	0.85	475.1	55.8	1.054	0.93
v3	4	2:2,3:0	2·2	1	0.2	18,153	109,950 ± 11,321	109,287	300,294	41,259	189,441	0.525	0.55	452.3	50.1	0.995	−0.17
v4	4	2:2,3:0	2·2	1	0.2	4,318	98,213 ± 11,891	89,084	284,920	30,600	175,609	0.618	0.85	477.4	46.6	0.889	−2.12
v5	6	2:1,3:1	2·3	1	1.2	6,702	120,439 ± 11,129	113,746	255,261	47,684	206,956	0.471	0.37	455.1	54.7	1.090	1.69
v6	6	2:1,3:1	2·3	1	1.3	9,999	106,987 ± 13,377	88,940	303,373	27,110	192,941	0.638	0.75	479.8	51.7	0.968	−0.59
v7	6	2:1,3:1	2·3	1	1.4	11,665	102,564 ± 11,181	96,354	300,067	39,065	165,278	0.556	1.01	478.4	49.8	0.928	−1.48
v11	8	2:3,3:0	2·2·2	1	0.3	6,800	106,131 ± 11,086	98,960	275,017	35,387	179,048	0.533	0.60	494.4	52.0	0.960	−0.86
v12	8	2:3,3:0	2·2·2	1	0.3	14,780	101,907 ± 10,828	92,979	272,998	40,331	183,359	0.542	0.83	497.0	49.7	0.922	−1.65
v16	8	2:3,3:0	2·2·2	1	0.3	1,645	119,943 ± 11,726	109,568	239,872	43,930	206,447	0.499	0.25	524.7	60.2	1.085	1.52
v17	8	2:3,3:0	2·2·2	1	0.3	7,372	115,632 ± 11,659	109,694	268,176	42,907	187,918	0.514	0.17	470.5	55.9	1.046	0.80
v8	9	2:0,3:2	3·3	1	0.3	10,219	109,017 ± 10,438	103,697	285,729	42,087	182,320	0.488	0.68	492.4	53.9	0.986	−0.36
v9	9	2:0,3:2	3·3	1	0.3	8,253	115,133 ± 11,577	100,237	296,521	53,583	196,127	0.513	0.75	465.5	53.1	1.042	0.72
v10	9	2:0,3:2	3·3	1	0.3	24,095	118,650 ± 11,243	115,195	290,262	50,501	196,404	0.483	0.44	469.0	54.6	1.074	1.36
v13	12	2:2,3:1	2·2·3	1	1.2	11,830	108,520 ± 10,968	97,379	234,447	38,476	184,881	0.516	0.23	467.2	51.4	0.982	−0.44
v14	12	2:2,3:1	2·2·3	1	1.3	12,866	116,114 ± 13,791	101,477	330,338	36,276	213,711	0.606	0.84	463.4	54.6	1.051	0.75
v15	12	2:2,3:1	2·2·3	1	1.4	4,396	107,111 ± 11,365	108,802	297,362	32,151	176,607	0.541	0.54	496.2	53.3	0.969	−0.67
v18	12	2:2,3:1	2·2·3	1	1.0	19,851	107,267 ± 9,327	100,444	278,261	48,470	172,965	0.444	0.63	493.1	53.0	0.971	−0.78
v19	12	2:2,3:1	2·2·3	1	1.1	6,023	112,031 ± 10,781	111,413	249,427	42,351	197,636	0.491	0.43	505.5	56.6	1.014	0.20
v20	12	2:2,3:1	2·2·3	1	1.3	4,407	108,347 ± 10,992	108,613	296,383	45,346	170,824	0.518	0.71	506.0	55.2	0.980	−0.47
v30	16	2:4,3:0	2·2·2·2	1	0.4	7,709	115,809 ± 11,396	115,898	295,369	45,963	197,577	0.502	0.70	489.1	56.6	1.048	0.85
v31	16	2:4,3:0	2·2·2·2	1	0.4	2,640	106,735 ± 12,310	94,681	315,041	34,985	194,660	0.588	0.87	482.0	52.1	0.966	−0.68
v35	16	2:4,3:0	2·2·2·2	1	0.4	8,246	107,498 ± 11,615	102,386	269,752	37,624	190,493	0.551	0.57	470.1	51.2	0.973	−0.59
v49	16	2:4,3:0	2·2·2·2	1	0.4	18,885	114,870 ± 10,543	108,745	290,170	51,516	185,697	0.468	0.55	450.9	52.9	1.039	0.74
v50	16	2:4,3:0	2·2·2·2	1	0.4	6,857	114,833 ± 11,074	108,087	307,120	45,414	185,621	0.492	0.59	497.7	56.0	1.039	0.70
v54	16	2:4,3:0	2·2·2·2	1	0.4	10,376	102,926 ± 11,012	99,180	241,473	28,912	190,701	0.546	0.42	505.3	51.2	0.931	−1.44
v21	18	2:1,3:2	2·3·3	1	2.9	11,455	112,883 ± 11,738	99,643	266,652	47,888	189,161	0.531	0.53	497.1	56.6	1.021	0.33
v22	18	2:1,3:2	2·3·3	1	3.0	15,974	109,559 ± 11,568	96,367	266,906	42,690	185,314	0.539	0.62	495.8	54.4	0.991	−0.24
v24	18	2:1,3:2	2·3·3	1	2.2	2,966	101,030 ± 10,864	94,179	233,492	26,473	174,056	0.549	0.21	511.8	52.5	0.914	−1.81
v25	18	2:1,3:2	2·3·3	1	2.3	3,805	107,351 ± 11,331	100,713	291,247	44,028	179,569	0.539	0.78	459.3	52.0	0.971	−0.63
v27	18	2:1,3:2	2·3·3	1	14.0	12,377	116,682 ± 12,463	111,643	283,637	39,516	204,337	0.545	0.51	496.9	69.7	1.056	0.91
v28	18	2:1,3:2	2·3·3	1	14.1	22,438	114,007 ± 11,852	101,854	307,333	51,244	194,447	0.530	1.00	499.6	69.4	1.032	0.51
v32	24	2:3,3:1	2·2·2·3	1	1.4	12,499	104,108 ± 11,467	107,472	288,113	36,527	174,317	0.562	0.79	480.5	49.4	0.942	−1.18
v34	24	2:3,3:1	2·2·2·3	1	1.6	15,858	108,564 ± 11,352	108,530	323,194	37,700	178,287	0.533	0.64	488.4	54.6	0.982	−0.42
v38	24	2:3,3:1	2·2·2·3	1	1.5	12,435	114,990 ± 13,251	104,090	333,980	34,564	204,166	0.588	0.77	505.0	57.9	1.040	0.61
v51	24	2:3,3:1	2·2·2·3	1	1.5	6,862	102,808 ± 11,532	87,064	257,480	37,974	184,400	0.572	0.72	518.1	53.9	0.930	−1.40

v53	24	2:3,3:1	2·2·2·3	1	1.5	19,780	124,391 ± 11,026	124,321	275,274	60,921	206,455	0.452	0.40	475.6	59.2	1.126	2.42
v57	24	2:3,3:1	2·2·2·3	1	1.9	13,533	112,062 ± 10,773	99,199	299,793	48,546	187,246	0.491	0.65	449.1	51.6	1.014	0.21
v40	36	2:2,3:2	2·2·3·3	1	2.2	12,413	124,879 ± 12,801	124,508	291,651	40,099	207,878	0.523	0.46	507.4	65.2	1.130	2.16
v43	36	2:2,3:2	2·2·3·3	1	3.0	21,773	107,911 ± 11,408	101,476	295,172	37,418	194,141	0.539	0.62	526.9	59.7	0.976	−0.53
v46	36	2:2,3:2	2·2·3·3	1	2.7	21,085	110,729 ± 11,322	106,115	296,805	45,855	177,070	0.522	0.80	470.9	54.1	1.002	−0.04
v59	36	2:2,3:2	2·2·3·3	1	2.2	5,569	106,115 ± 11,763	102,334	264,839	31,099	198,190	0.566	0.58	487.6	52.6	0.960	−0.81
v62	36	2:2,3:2	2·2·3·3	1	2.5	7,906	122,004 ± 12,078	117,404	274,274	35,541	211,944	0.505	0.42	484.5	60.9	1.104	1.82
v65	36	2:2,3:2	2·2·3·3	1	3.1	8,182	110,603 ± 11,721	105,609	390,832	44,186	174,462	0.541	1.30	490.3	55.9	1.001	−0.06

B1: every measured curve, one instance, 100 paired seeds per arm

Reference √(πn/2) = 328,549 (×1.016 for the 32-adding walk: 333,806); 4600 solves, all returned the planted k; rho wall 217.7 s on 4 threads.

arm	kind	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	mean ms	e2e ms	ratio to √(πn/2)	z vs rest
all isogenous	pooled	40	4000	3,097	334,995 ± 5,546	315,171	1,194,318	0.534	186.9		1.020	
controls	iso / representation	3	300	12,380	326,101 ± 18,655	310,075	849,215	0.506	183.8		0.993	
rep0	iso-control	1	100	62,580	342,528 ± 30,807	322,683	750,919	0.459	198.7	198.7	1.043 [0.955, 1.139]	0.47
rep1	iso-control	1	100	31,922	294,558 ± 30,640	297,517	683,137	0.531	173.9	173.9	0.897 [0.810, 0.991]	−2.55
rep2	iso-control	1	100	12,380	341,215 ± 34,746	314,338	849,215	0.520	178.7	178.7	1.039 [0.940, 1.148]	0.35
aa	aa-replica	1	100	38,487	381,345 ± 40,342	376,634	883,836	0.540	217.0	217.0	1.161 [1.049, 1.285]	2.23
plant-neg	plant (expected 0.707)	1	100	44,958	253,302 ± 24,392	251,460	602,194	0.491	146.7	146.7	0.771 [0.702, 0.847]	−6.40
plant-r4	plant (expected 1.137)	1	100	14,574	427,094 ± 42,586	390,347	1,102,121	0.509	255.8	255.8	1.300 [1.177, 1.435]	4.20

Pooled by endomorphism-ring stratum (conductor of End relative to the walked primes), by level in each volcano, by distance from the root, by the degree and kernel field of the edge that reached the curve:

group	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	ratio to √(πn/2)	z vs rest
conductor:1	1	100	35,288	366,663 ± 35,602	377,179	1,024,161	0.495	1.116	1.77
conductor:119	10	1000	5,455	335,870 ± 11,292	319,961	1,110,313	0.542	1.022	0.18
conductor:17	4	400	11,822	339,940 ± 17,029	317,951	851,524	0.511	1.035	0.60
conductor:49	8	800	24,662	338,987 ± 12,703	313,229	1,023,441	0.541	1.032	0.69
conductor:7	6	600	3,097	326,540 ± 14,163	306,489	1,014,590	0.542	0.994	−1.27
conductor:833	11	1100	19,041	331,229 ± 10,365	310,903	1,194,318	0.529	1.008	−0.83
dist:0	1	100	22,789	324,778 ± 32,038	301,947	815,413	0.503	0.989	−0.63
dist:1	5	500	20,622	341,887 ± 15,336	334,944	953,365	0.512	1.041	0.94
dist:2	8	800	5,455	325,264 ± 12,690	307,207	1,194,318	0.563	0.990	−1.69
dist:3	26	2600	3,097	337,056 ± 6,879	315,377	1,110,313	0.531	1.026	0.99
ell17:level0	15	1500	3,097	335,853 ± 9,153	312,357	1,024,161	0.539	1.022	0.23
ell17:level1	25	2500	5,455	334,479 ± 6,971	316,248	1,194,318	0.532	1.018	−0.23
ell7:level0	5	500	11,822	345,285 ± 15,384	323,644	1,024,161	0.508	1.051	1.40
ell7:level1	16	1600	3,097	332,371 ± 8,833	315,366	1,110,313	0.542	1.012	−0.76
ell7:level2	19	1900	19,041	334,496 ± 8,038	310,903	1,194,318	0.534	1.018	−0.17
field_degree:11	4	400	5,455	323,792 ± 17,189	305,641	953,365	0.542	0.986	−1.34
field_degree:3	8	800	11,822	336,561 ± 11,944	317,951	858,003	0.512	1.024	0.28
field_degree:5	3	300	26,890	340,246 ± 20,444	351,385	983,254	0.531	1.036	0.52
field_degree:8	24	2400	3,097	336,109 ± 7,287	313,414	1,194,318	0.542	1.023	0.49
parent_ell:17	24	2400	3,097	336,109 ± 7,287	313,414	1,194,318	0.542	1.023	0.49
parent_ell:23	4	400	5,455	323,792 ± 17,189	305,641	953,365	0.542	0.986	−1.34
parent_ell:31	3	300	26,890	340,246 ± 20,444	351,385	983,254	0.531	1.036	0.52
parent_ell:7	8	800	11,822	336,561 ± 11,944	317,951	858,003	0.512	1.024	0.28

Across the 40 measured curves: Kruskal–Wallis H = 47.79 (df 39, p = 0.158); ANOVA F = 1.112 (p = 0.292); between-curve variance ratio 1.112 (observed sd of curve means 18,861 against 17,896 expected from the pooled single-solve spread); largest |z| 3.02 against the Bonferroni threshold 3.23; curves beyond it: none.

Per curve (iterations = group operations; d = kernel field degree of the edge that reached it; transport = ms to find and apply the isogenies from the root):

curve	stratum f	levels	path (ℓ)	d	transport ms	min	mean ± CI	median	max	p10	p90	sd/mean	skew	ns/op	e2e ms	ratio	z
v49	1	7:0,17:0	7·7·17	8	142.6	35,288	366,663 ± 35,602	377,179	1,024,161	154,528	552,766	0.495	0.96	582.5	350.2	1.116	1.77
v13	7	7:1,17:0	7·17	8	81.1	21,719	343,183 ± 37,034	332,185	1,014,590	103,458	580,834	0.551	0.58	571.4	273.5	1.045	0.44
v62	7	7:1,17:0	7·17·17	8	651.1	34,362	333,524 ± 38,817	310,484	923,645	120,703	628,072	0.594	0.71	537.1	833.5	1.015	−0.08
v66	7	7:1,17:0	7·17·17	8	651.0	14,692	316,284 ± 37,382	285,249	945,638	95,778	564,833	0.603	0.97	542.0	818.6	0.963	−1.00
v75	7	7:1,17:0	7·23·17	8	1,602.3	50,490	291,128 ± 28,641	265,758	741,221	115,538	497,951	0.502	0.76	557.8	1,760.8	0.886	−3.02
v83	7	7:1,17:0	7·23·17	8	1,599.2	16,171	348,454 ± 31,424	341,908	834,878	137,114	520,791	0.460	0.48	577.5	1,795.4	1.061	0.85
v91	7	7:1,17:0	7·31·17	8	2,244.1	3,097	326,669 ± 33,388	312,357	782,909	106,703	574,033	0.521	0.27	587.4	2,431.7	0.994	−0.49
v11	17	7:0,17:1	7·7	3	3.9	11,822	335,015 ± 33,776	309,387	802,651	143,651	574,158	0.514	0.49	584.4	198.7	1.020	0.00
v72	17	7:0,17:1	7·23·7	3	1,441.8	29,513	337,621 ± 35,385	318,559	851,524	115,462	574,682	0.535	0.73	562.5	1,629.7	1.028	0.15
v79	17	7:0,17:1	7·23·7	3	1,446.0	22,492	334,058 ± 32,387	318,872	830,547	139,303	530,185	0.495	0.56	569.4	1,638.9	1.017	−0.06
v87	17	7:0,17:1	7·31·7	3	2,121.2	71,655	353,066 ± 34,982	323,644	824,689	128,165	575,430	0.506	0.57	566.1	2,323.8	1.075	1.03
v2	49	7:2,17:0	17	8	81.5	44,486	339,175 ± 36,279	317,051	876,232	131,223	600,591	0.546	0.59	582.6	276.6	1.032	0.23
v26	49	7:2,17:0	17·17	8	482.2	39,924	308,767 ± 36,377	265,044	912,260	99,901	562,159	0.601	0.91	578.7	664.8	0.940	−1.43
v36	49	7:2,17:0	17·31	5	2,709.8	36,251	355,596 ± 37,628	355,821	983,254	133,629	583,939	0.540	0.67	554.6	2,900.8	1.082	1.09
v38	49	7:2,17:0	7·7·17	8	193.8	42,680	344,832 ± 32,220	335,743	773,518	167,099	562,203	0.477	0.54	542.2	380.0	1.050	0.60
v41	49	7:2,17:0	7·7·17	8	111.6	46,373	365,585 ± 38,191	345,317	1,023,441	137,624	616,490	0.533	0.73	543.9	312.3	1.113	1.59
v93	49	7:2,17:0	17·17·17	8	1,042.6	29,591	305,383 ± 33,956	289,685	897,535	112,504	526,114	0.567	0.96	599.6	1,223.8	0.929	−1.73
v140	49	7:2,17:0	17·23·17	8	2,216.8	24,662	335,485 ± 36,640	284,318	785,191	107,672	626,215	0.557	0.59	576.4	2,401.6	1.021	0.03
v155	49	7:2,17:0	17·23·17	8	2,582.9	49,341	357,073 ± 35,200	343,991	885,343	137,199	610,919	0.503	0.40	584.7	2,791.1	1.087	1.25
v1	119	7:1,17:1	7	3	1.0	20,622	353,067 ± 36,602	367,182	857,753	134,591	562,774	0.529	0.34	558.0	194.1	1.075	0.98
v14	119	7:1,17:1	7·23	11	1,438.4	44,750	318,712 ± 35,373	307,207	789,607	100,598	569,934	0.566	0.61	535.7	1,605.4	0.970	−0.91
v15	119	7:1,17:1	7·23	11	1,438.4	5,455	301,465 ± 34,853	268,856	833,554	105,210	549,533	0.590	0.94	550.8	1,601.0	0.918	−1.91
v16	119	7:1,17:1	7·31	5	2,113.3	26,890	322,942 ± 36,705	334,478	858,852	94,775	555,661	0.580	0.42	550.2	2,297.2	0.983	−0.65
v44	119	7:1,17:1	7·7·7	3	7.9	17,166	321,699 ± 30,800	315,112	787,011	150,913	538,173	0.488	0.35	584.2	192.8	0.979	−0.85
v47	119	7:1,17:1	7·7·7	3	7.9	33,525	339,352 ± 35,150	317,374	858,003	112,541	542,207	0.528	0.66	552.0	196.1	1.033	0.25
v53	119	7:1,17:1	7·17·17	8	651.1	5,617	373,320 ± 38,762	358,260	842,935	131,185	667,091	0.530	0.49	554.3	858.9	1.136	1.97
v56	119	7:1,17:1	7·17·17	8	651.1	38,445	339,586 ± 39,775	312,970	1,110,313	125,341	615,612	0.598	1.28	543.1	834.6	1.034	0.23
v59	119	7:1,17:1	7·17·17	8	651.0	61,156	356,817 ± 34,550	356,914	773,643	127,421	565,926	0.494	0.27	567.0	852.9	1.086	1.25
v63	119	7:1,17:1	7·17·17	8	651.0	54,281	331,740 ± 33,109	323,413	884,796	126,947	551,758	0.509	0.80	550.9	835.7	1.010	−0.19
v0	833	7:2,17:1	root	—	0.0	22,789	324,778 ± 32,038	301,947	815,413	120,711	557,670	0.503	0.56	565.6	185.2	0.989	−0.63
v3	833	7:2,17:1	23	11	1,641.5	31,888	318,429 ± 34,205	297,665	953,365	120,048	535,554	0.548	0.93	583.3	1,822.1	0.969	−0.96
v4	833	7:2,17:1	23	11	1,641.5	58,567	356,561 ± 32,618	347,989	773,229	120,467	576,863	0.467	0.25	540.7	1,829.1	1.085	1.31
v5	833	7:2,17:1	31	5	1,877.7	70,078	342,201 ± 31,670	366,783	736,067	130,987	554,775	0.472	0.17	590.2	2,070.3	1.042	0.45
v22	833	7:2,17:1	17·17	8	482.3	26,864	316,434 ± 35,254	300,516	1,194,318	134,766	542,432	0.568	1.52	538.4	657.1	0.963	−1.05
v74	833	7:2,17:1	7·23·7	3	1,441.9	22,991	318,607 ± 31,279	313,141	774,883	135,688	523,633	0.501	0.50	549.4	1,613.1	0.970	−1.04
v95	833	7:2,17:1	17·17·17	8	1,042.6	22,666	337,997 ± 34,303	309,924	968,233	142,358	579,774	0.518	0.81	568.1	1,236.3	1.029	0.17
v110	833	7:2,17:1	17·17·17	8	986.0	19,041	338,832 ± 34,684	312,236	773,387	136,441	586,951	0.522	0.47	547.5	1,169.9	1.031	0.22
v126	833	7:2,17:1	17·23·17	8	2,216.8	21,176	327,879 ± 37,793	310,771	892,390	127,592	582,171	0.588	0.69	528.2	2,388.9	0.998	−0.37
v143	833	7:2,17:1	17·23·17	8	2,582.9	23,635	311,346 ± 32,760	295,985	792,908	122,815	538,890	0.537	0.60	568.6	2,758.8	0.948	−1.43
v160	833	7:2,17:1	17·31·17	8	3,125.5	56,237	350,458 ± 40,814	327,353	889,837	111,942	645,664	0.594	0.71	573.6	3,325.3	1.067	0.75

K: every measured curve, one instance, 100 paired seeds per arm

Reference √(πn/2) = 929,277 (×1.016 for the 32-adding walk: 944,146); 1900 solves, all returned the planted k; rho wall 218.1 s on 4 threads.

arm	kind	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	mean ms	e2e ms	ratio to √(πn/2)	z vs rest
all isogenous	pooled	11	1100	18,623	924,231 ± 28,590	875,347	2,748,554	0.523	468.1		0.995	
controls	iso / representation	3	300	49,568	921,314 ± 58,395	850,882	2,273,263	0.560	519.7		0.991	
rep0	iso-control	1	100	124,818	913,447 ± 103,488	850,003	2,213,473	0.578	456.7	456.7	0.983 [0.880, 1.097]	−0.20
rep1	iso-control	1	100	195,254	955,959 ± 98,363	864,308	2,273,263	0.525	477.4	477.4	1.029 [0.931, 1.139]	0.61
rep2	iso-control	1	100	49,568	894,534 ± 102,151	845,960	2,246,633	0.583	625.1	625.1	0.963 [0.861, 1.076]	−0.55

aa	aa-replica	1	100	39,408	901,192 ± 96,770	888,631	2,299,438	0.548	456.2	456.2	0.970 [0.875, 1.078]	−0.45
plant-neg	plant (expected 0.707)	1	100	71,980	714,180 ± 63,358	701,560	1,617,133	0.453	374.5	374.5	0.769 [0.705, 0.838]	−5.92
plant-r4	plant (expected 1.137)	1	100	164,295	1,261,813 ± 136,603	1,237,400	3,741,923	0.552	649.5	649.5	1.358 [1.221, 1.508]	4.74
fold-frob	fold (expected 0.156)	1	100	10,324	143,860 ± 13,790	131,211	373,676	0.489	274.9	274.9	0.155 [0.141, 0.170]	−48.19
fold-frobneg	fold (expected 0.110)	1	100	5,481	105,399 ± 9,888	102,279	228,127	0.479	212.3	212.3	0.113 [0.103, 0.124]	−53.05

Pooled by endomorphism-ring stratum (conductor of End relative to the walked primes), by level in each volcano, by distance from the root, by the degree and kernel field of the edge that reached the curve:

group	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	ratio to √(πn/2)	z vs rest
conductor:1	1	100	214,768	925,603 ± 100,645	827,699	2,459,365	0.555	0.996	0.03
conductor:1721	3	300	69,369	930,450 ± 55,513	874,470	2,681,756	0.527	1.001	0.26
conductor:409	3	300	23,089	938,027 ± 54,747	885,274	2,568,833	0.516	1.009	0.58
conductor:703889	4	400	18,623	908,875 ± 46,286	885,459	2,748,554	0.520	0.978	−0.80
dist:0	1	100	214,768	925,603 ± 100,645	827,699	2,459,365	0.555	0.996	0.03
dist:1	6	600	23,089	934,239 ± 38,952	876,383	2,681,756	0.521	1.005	0.75
dist:2	4	400	18,623	908,875 ± 46,286	885,459	2,748,554	0.520	0.978	−0.80
ell1721:level0	4	400	23,089	934,921 ± 48,095	861,623	2,568,833	0.525	1.006	0.55
ell1721:level1	7	700	18,623	918,122 ± 35,558	877,425	2,748,554	0.523	0.988	−0.55
ell409:level0	4	400	69,369	929,238 ± 48,584	851,085	2,681,756	0.534	1.000	0.26
ell409:level1	7	700	18,623	921,369 ± 35,347	885,274	2,748,554	0.518	0.991	−0.26
field_degree:204	5	500	23,089	936,181 ± 41,457	899,512	2,568,833	0.505	1.007	0.75
field_degree:215	5	500	18,623	912,005 ± 42,867	851,085	2,748,554	0.536	0.981	−0.76
parent_ell:1721	5	500	18,623	912,005 ± 42,867	851,085	2,748,554	0.536	0.981	−0.76
parent_ell:409	5	500	23,089	936,181 ± 41,457	899,512	2,568,833	0.505	1.007	0.75

Across the 11 measured curves: Kruskal–Wallis H = 13.27 (df 10, p = 0.209); ANOVA F = 1.699 (p = 0.076); between-curve variance ratio 1.699 (observed sd of curve means 62,858 against 48,378 expected from the pooled single-solve spread); largest |z| 2.38 against the Bonferroni threshold 2.84; curves beyond it: none.

Per curve (iterations = group operations; d = kernel field degree of the edge that reached it; transport = ms to find and apply the isogenies from the root):

curve	stratum f	levels	path (ℓ)	d	transport ms	min	mean ± CI	median	max	p10	p90	sd/mean	skew	ns/op	e2e ms	ratio	z
v0	1	409:0,1721:0	root	—	0.0	214,768	925,603 ± 100,645	827,699	2,459,365	364,498	1,728,974	0.555	0.96	503.3	458.8	0.996	0.03
v1	409	409:1,1721:0	409	204	177,777.3	23,089	854,793 ± 86,269	913,111	2,180,110	281,300	1,439,885	0.515	0.20	507.5	178,213.3	0.920	−1.64
v2	409	409:1,1721:0	409	204	177,777.2	98,627	1,047,479 ± 107,530	911,741	2,568,833	423,555	1,852,848	0.524	0.79	506.8	178,307.9	1.127	2.38
v3	409	409:1,1721:0	409	204	177,777.2	166,837	911,810 ± 85,974	836,253	2,321,513	388,805	1,548,816	0.481	0.72	507.1	178,238.0	0.981	−0.29
v4	1721	409:0,1721:1	1721	215	674,886.9	109,036	945,704 ± 89,001	887,590	2,473,027	389,168	1,513,127	0.480	0.72	503.5	675,360.2	1.018	0.49
v5	1721	409:0,1721:1	1721	215	674,887.0	69,369	883,607 ± 88,336	828,617	2,255,262	402,566	1,556,484	0.510	0.58	511.0	675,338.2	0.951	−0.94
v6	1721	409:0,1721:1	1721	215	674,886.9	104,232	962,040 ± 109,839	885,636	2,681,756	379,429	1,716,502	0.583	0.74	523.5	675,386.6	1.035	0.72
v7	703889	409:1,1721:1	409·1721	215	683,386.1	140,865	947,168 ± 97,043	958,036	2,748,554	362,800	1,415,980	0.523	0.75	505.2	683,865.4	1.019	0.49
v11	703889	409:1,1721:1	409·1721	215	536,149.5	18,623	821,508 ± 92,549	747,495	2,337,890	273,799	1,426,089	0.575	0.83	501.1	536,557.4	0.884	−2.28
v16	703889	409:1,1721:1	1721·409	204	818,372.1	72,509	885,687 ± 80,091	877,425	2,133,616	380,598	1,373,108	0.461	0.47	506.6	818,822.0	0.953	−0.97
v20	703889	409:1,1721:1	1721·409	204	890,492.5	130,052	981,138 ± 97,796	931,889	2,514,052	343,773	1,612,007	0.509	0.29	511.8	890,994.5	1.056	1.20

P1: 100 generators of ⟨P⟩ per curve, the same secret, 25 paired
seeds each

curve	generators	solves	mean ± CI	min / max of generator means	sd of generator means (expected)	variance ratio	KW p	ANOVA p	max	z
v0	100	2500	209,581 ± 4,287	144,169 / 254,175	21,116 (21,873)	0.929	0.520	0.675	3.37 (3.93)	0.08 / 0.12 / 0.12 / 0.08
v98	100	2500	206,581 ± 4,208	151,961 / 267,382	21,863 (21,472)	1.038	0.454	0.380	3.76 (3.93)	0.09 / 0.09 / −0.02 / 0.01

v727	100	2500	205,424 ± 4,242	158,417 / 271,380	21,202 (21,641)	0.958	0.594	0.598	2.75 (3.93)	−0.02 / 0.00 / −0.01 / 0.10
v1113	100	2500	204,670 ± 4,222	153,875 / 252,148	19,852 (21,543)	0.844	0.918	0.863	3.00 (3.93)	−0.03 / −0.03 / 0.12 / −0.03
v1646	100	2500	201,327 ± 4,189	157,103 / 264,191	19,991 (21,375)	0.870	0.676	0.814	2.74 (3.93)	−0.09 / −0.10 / 0.09 / 0.09
v2266	100	2500	206,620 ± 4,170	159,556 / 266,249	21,380 (21,275)	1.010	0.441	0.454	3.07 (3.93)	0.02 / 0.03 / −0.03 / 0.00

All generator arms together: KW p = 0.775, variance ratio 0.947, flagged arms (family 600): none; pooled mean 205,700 = 1.015 × √(πn/2); 15000 solves, all correct.

P0: 100 generators of ⟨P⟩ per curve, the same secret, 25 paired
seeds each

curve	generators	solves	mean ± CI	min / max of generator means	sd of generator means (expected)	variance ratio	KW p	ANOVA p	max	z
v0	100	2500	111,320 ± 2,226	80,561 / 143,142	12,707 (11,359)	1.265	0.020	0.043	3.77 (3.84)	0.13 / 0.13 / 0.03 / 0.03
v11	100	2500	111,220 ± 2,301	85,983 / 151,042	10,627 (11,740)	0.813	0.931	0.909	2.87 (3.84)	−0.01 / 0.03 / −0.01 / 0.07
v22	100	2500	113,246 ± 2,280	82,190 / 149,754	10,536 (11,630)	0.815	0.884	0.907	3.23 (3.84)	−0.05 / −0.03 / −0.06 / 0.00
v40	100	2500	114,392 ± 2,373	89,771 / 145,759	11,076 (12,110)	0.831	0.936	0.884	2.65 (3.84)	0.25 / 0.17 / 0.19 / −0.06

All generator arms together: KW p = 0.818, variance ratio 0.929, flagged arms (family 400): none; pooled mean 112,545 = 1.018 × √(πn/2); 10000 solves, all correct.

B1: 64 generators of ⟨P⟩ per curve, the same secret, 25 paired seeds
each

curve	generators	solves	mean ± CI	min / max of generator means	sd of generator means (expected)	variance ratio	KW p	ANOVA p	max	z
v0	64	1600	339,248 ± 8,456	222,426 / 434,482	35,684 (34,514)	1.072	0.185	0.329	4.53 (3.83)	−0.00 / 0.03 / −0.09 / 0.06
v11	64	1600	340,907 ± 8,832	276,194 / 419,005	37,752 (36,047)	1.101	0.187	0.276	3.23 (3.83)	0.05 / 0.07 / −0.14 / 0.27
v36	64	1600	327,748 ± 8,259	244,025 / 388,084	35,851 (33,712)	1.137	0.121	0.219	3.41 (3.83)	−0.14 / −0.12 / −0.07 / 0.18
v56	64	1600	333,103 ± 8,473	242,652 / 433,712	36,991 (34,583)	1.151	0.424	0.199	3.17 (3.83)	−0.04 / −0.07 / 0.04 / −0.04
v74	64	1600	333,929 ± 8,324	264,611 / 423,173	35,124 (33,975)	1.072	0.284	0.330	2.30 (3.83)	−0.10 / −0.09 / 0.07 / 0.15
v95	64	1600	324,744 ± 8,325	252,038 / 394,414	35,424 (33,981)	1.091	0.286	0.295	2.78 (3.83)	−0.18 / −0.24 / −0.22 / −0.02

All generator arms together: KW p = 0.039, variance ratio 1.117, flagged arms (family 384): ['v0g59']; pooled mean 333,280 = 1.014 × √(πn/2); 9600 solves, all correct.

K: 64 generators of ⟨P⟩ per curve, the same secret, 25 paired seeds
each

curve	generators	solves	mean ± CI	min / max of generator means	sd of generator means (expected)	variance ratio	KW p	ANOVA p	max	z
v0	64	1600	940,718 ± 24,311	739,839 / 1,257,941	101,307 (99,230)	1.044	0.456	0.384	2.87 (3.65)	0.08 / 0.01 / 0.04 / 0.03
v3	64	1600	942,237 ± 25,042	708,728 / 1,331,359	102,029 (102,210)	0.996	0.724	0.486	3.37 (3.65)	0.02 / 0.02 / −0.18 / 0.12
v7	64	1600	939,088 ± 23,310	704,508 / 1,158,267	94,978 (95,141)	0.996	0.338	0.486	3.52 (3.65)	−0.12 / −0.02 / −0.20 / 0.03

All generator arms together: KW p = 0.584, variance ratio 1.002, flagged arms (family 192): none; pooled mean 940,681 = 1.012 × √(πn/2); 4800 solves, all correct.

P1: instances drawn on 5 source curves, carried to every measured
curve, 25 paired seeds per cell

source	drawn on	targets	solves	mean ± CI	ratio	z vs other sources
0	v0	104	2600	208,513 ± 4,138	1.029	1.33
1	v1	104	2600	204,614 ± 4,204	1.010	−0.73
2	v487	104	2600	204,109 ± 4,107	1.007	−1.01
3	v1113	104	2600	205,421 ± 3,955	1.014	−0.32
4	v2004	104	2600	207,355 ± 4,156	1.023	0.72

Two-way decomposition of the 13000 solves: η^2 (target curve) = 0.0085, η^2 (source curve) = 0.0002; Kruskal–Wallis by target p = 0.191 (df 103), by source p = 0.367 (df 4); flagged cells (family 520): ['s1v1343', 's4v625']; all solves correct.

P0: instances drawn on 4 source curves, carried to every measured
curve, 25 paired seeds per cell

source	drawn on	targets	solves	mean ± CI	ratio	z vs other sources
0	v0	45	1125	112,615 ± 3,494	1.019	−0.01
1	v1	45	1125	115,419 ± 3,378	1.044	1.86

2	v15	45	1125	109,743 ± 3,373	0.993	−1.93
3	v34	45	1125	112,763 ± 3,387	1.020	0.09

Two-way decomposition of the 4500 solves: η^2 (target curve) = 0.0112, η^2 (source curve) = 0.0012; Kruskal–Wallis by target p = 0.223 (df 44), by source p = 0.130 (df 3); flagged cells (family 180): ['s0v53']; all solves correct.

B1: instances drawn on 5 source curves, carried to every measured
curve, 25 paired seeds per cell

source	drawn on	targets	solves	mean ± CI	ratio	z vs other sources
0	v0	40	1000	329,998 ± 10,940	1.004	−0.91
1	v1	40	1000	332,260 ± 10,862	1.011	−0.46
2	v16	40	1000	337,637 ± 10,476	1.028	0.64
3	v56	40	1000	337,887 ± 11,081	1.028	0.67
4	v87	40	1000	334,922 ± 10,569	1.019	0.08

Two-way decomposition of the 5000 solves: η^2 (target curve) = 0.0089, η^2 (source curve) = 0.0003; Kruskal–Wallis by target p = 0.152 (df 39), by source p = 0.562 (df 4); flagged cells (family 200): none; all solves correct.

K: instances drawn on 3 source curves, carried to every measured
curve, 25 paired seeds per cell

source	drawn on	targets	solves	mean ± CI	ratio	z vs other sources
0	v0	11	275	910,047 ± 58,793	0.979	−0.31
1	v1	11	275	936,806 ± 56,600	1.008	0.81
2	v6	11	275	905,749 ± 58,764	0.975	−0.48

Two-way decomposition of the 825 solves: η^2 (target curve) = 0.0158, η^2 (source curve) = 0.0008; Kruskal–Wallis by target p = 0.357 (df 10), by source p = 0.485 (df 2); flagged cells (family 33): none; all solves correct.

Figures
Every figure draws recorded values only (`plot.py`). The standard volcano diagrams (`tools/volcano-diagram`) place each measured curve at its level in one ℓ-volcano with every walked edge drawn; the 3D heat maps stack the endomorphism-ring strata (rings, crater on top) and colour each measured curve by its mean operations or its mean end-to-end time (transport + solve).

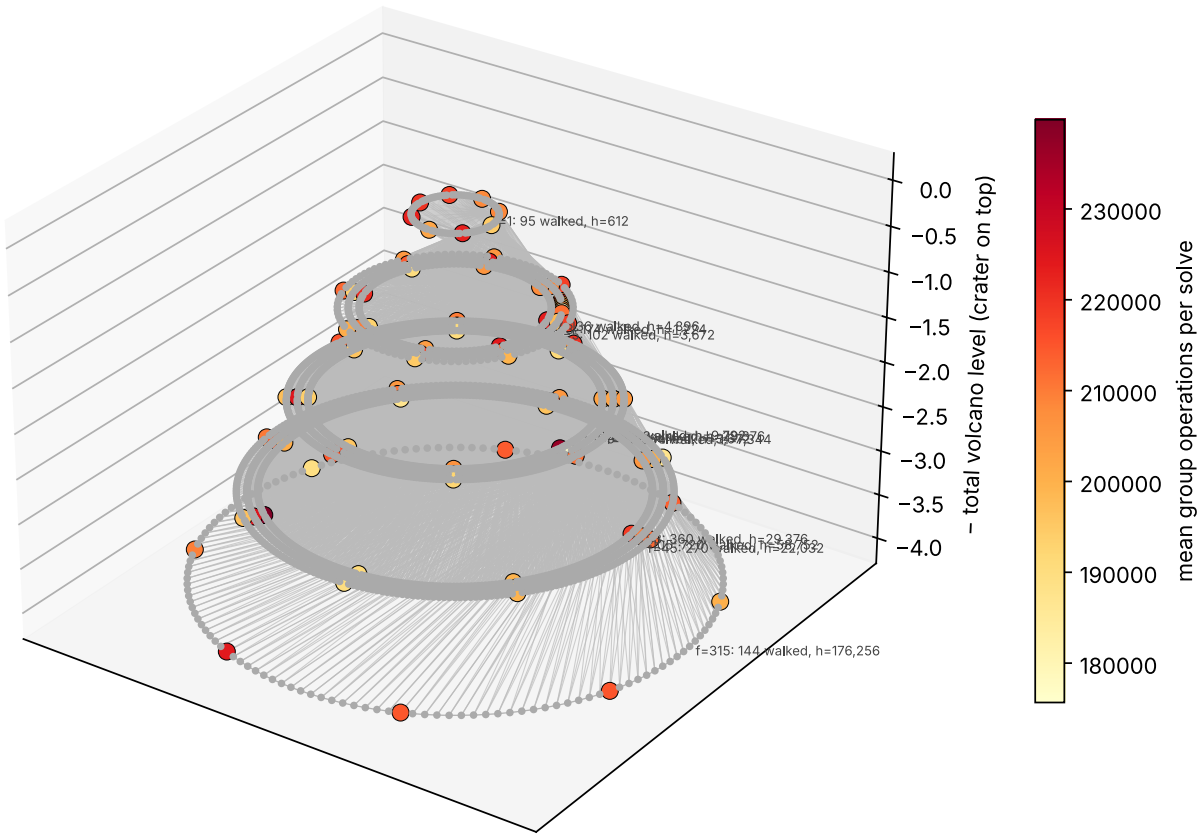


Figure 1: P1 3D class heat map, mean group operations per solve

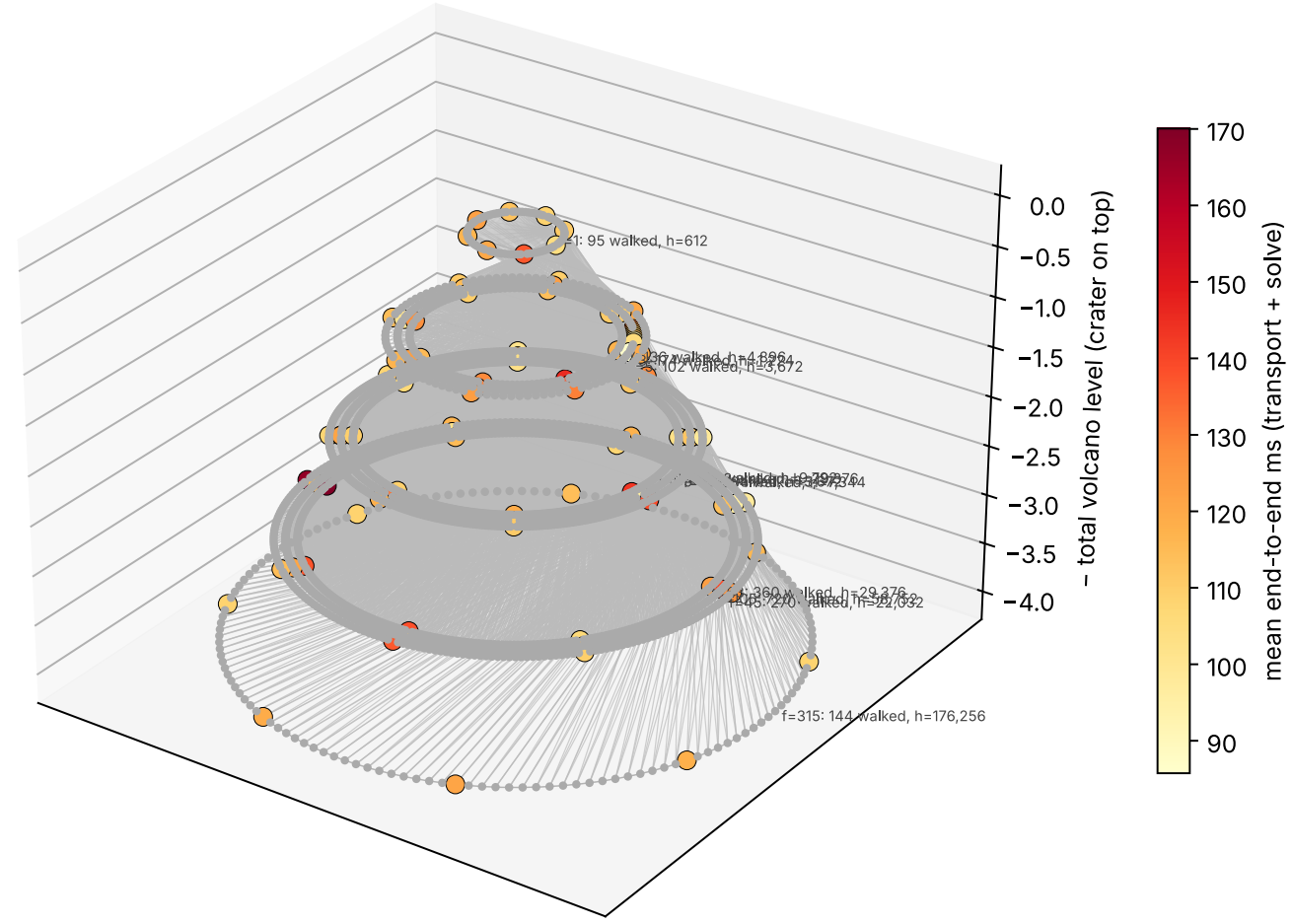


Figure 2: P1 3D class heat map, mean end-to-end ms (transport + solve)

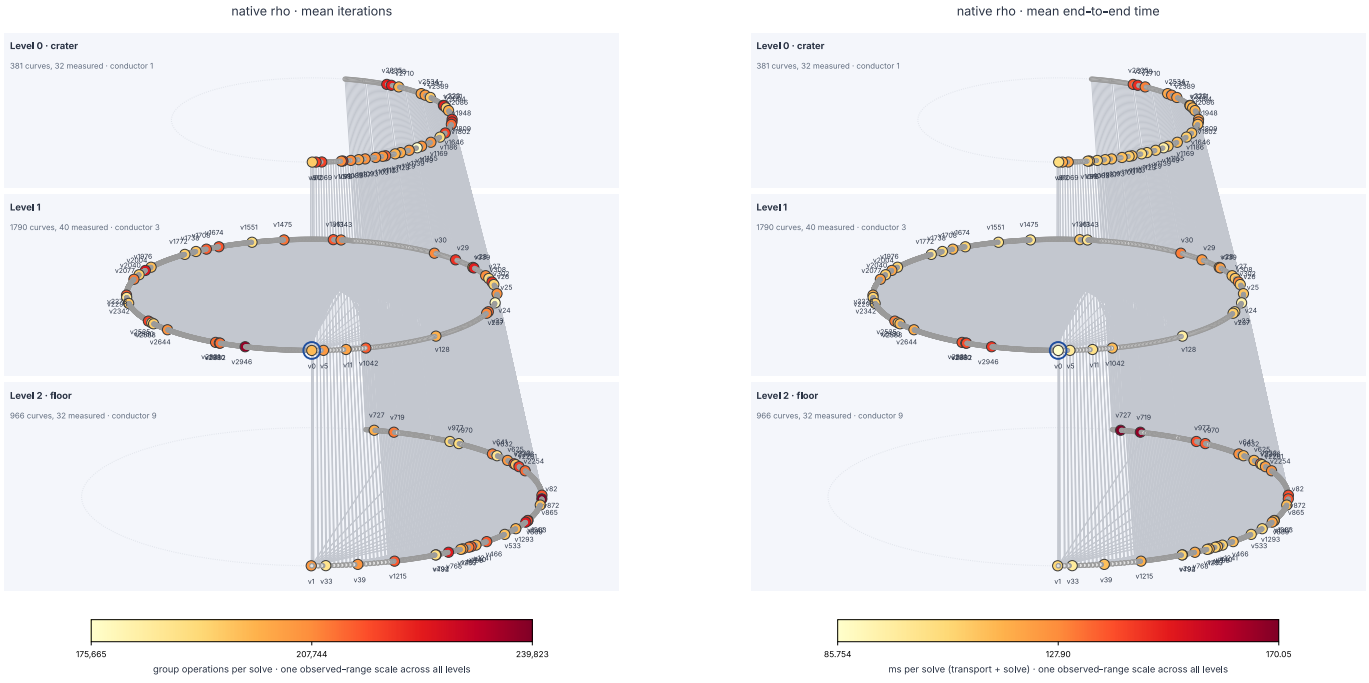


Figure 3: P1 volcano-ell3: standard volcano diagram

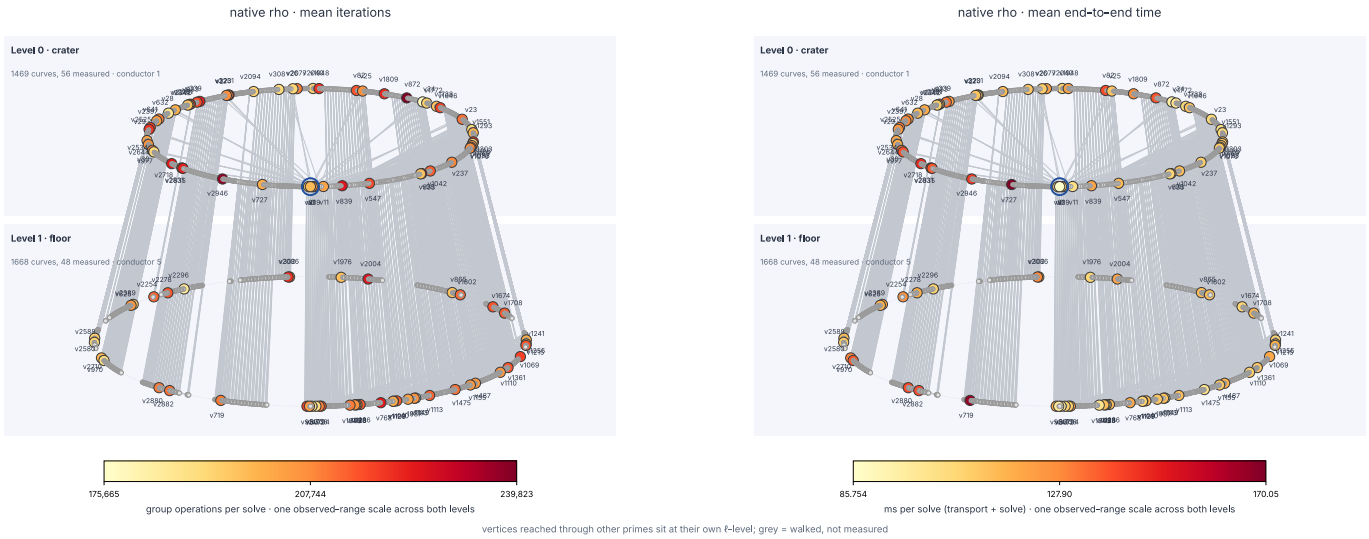


Figure 4: P1 volcano-ell5: standard volcano diagram

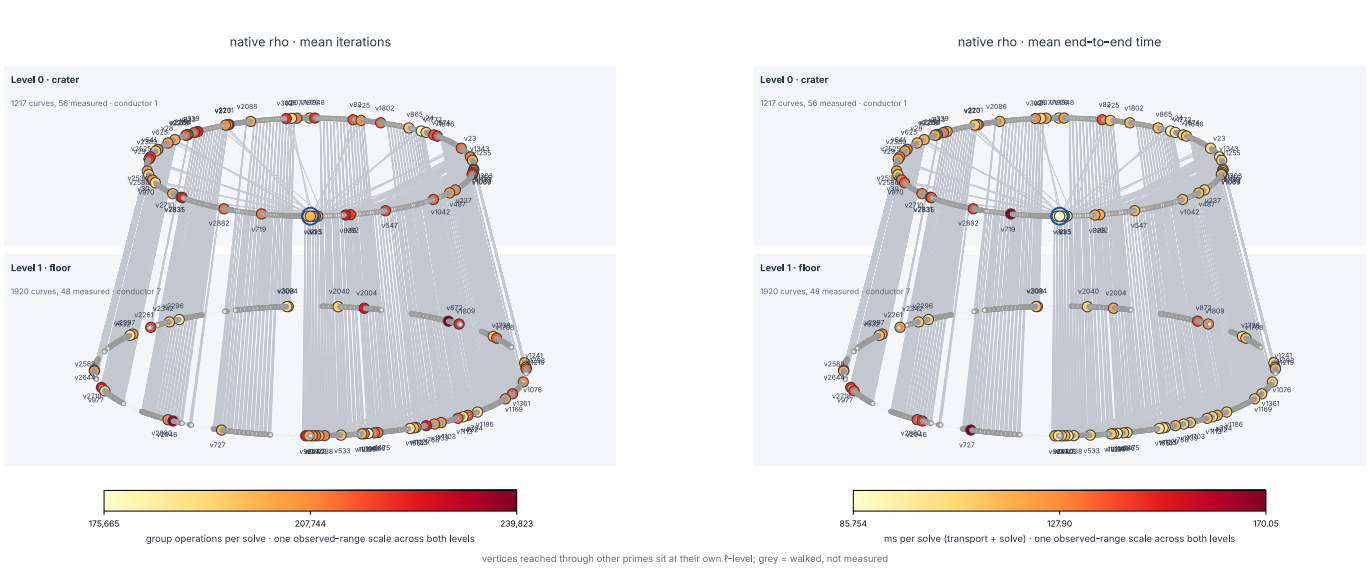


Figure 5: P1 volcano-ell7: standard volcano diagram

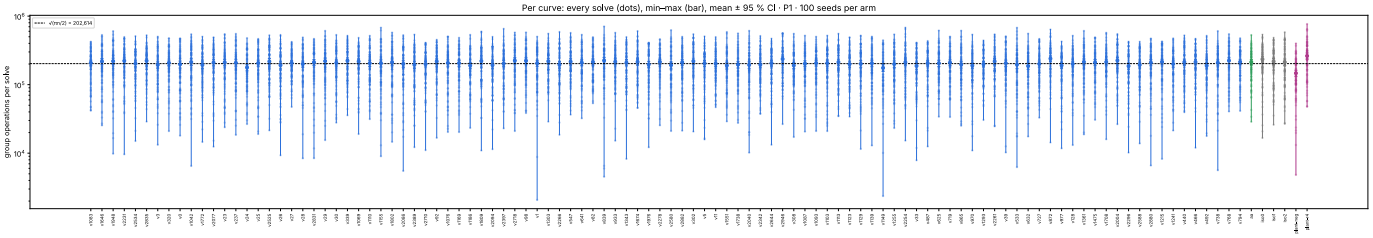


Figure 6: P1 every solve, min–max, mean ± CI per arm

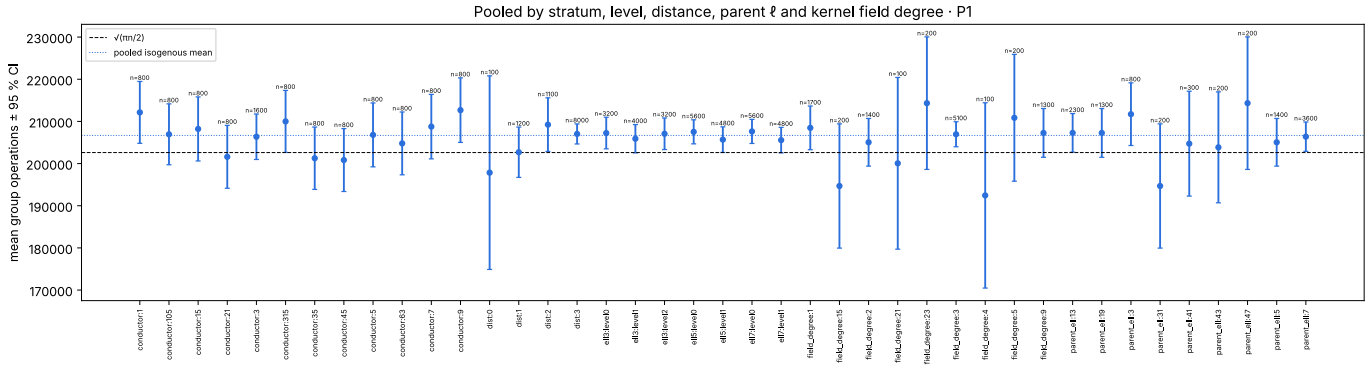


Figure 7: P1 pooled groups

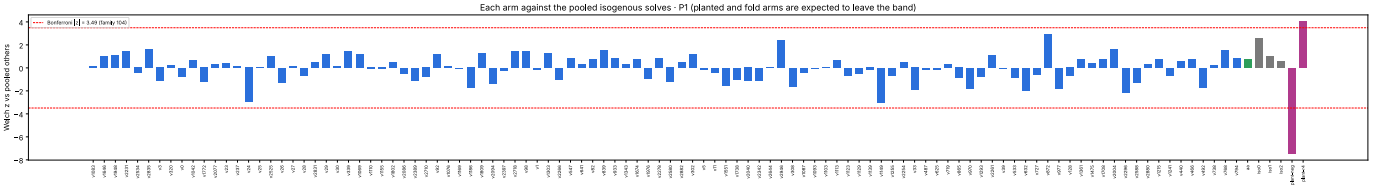


Figure 8: P1 each arm against the pooled isogenous solves

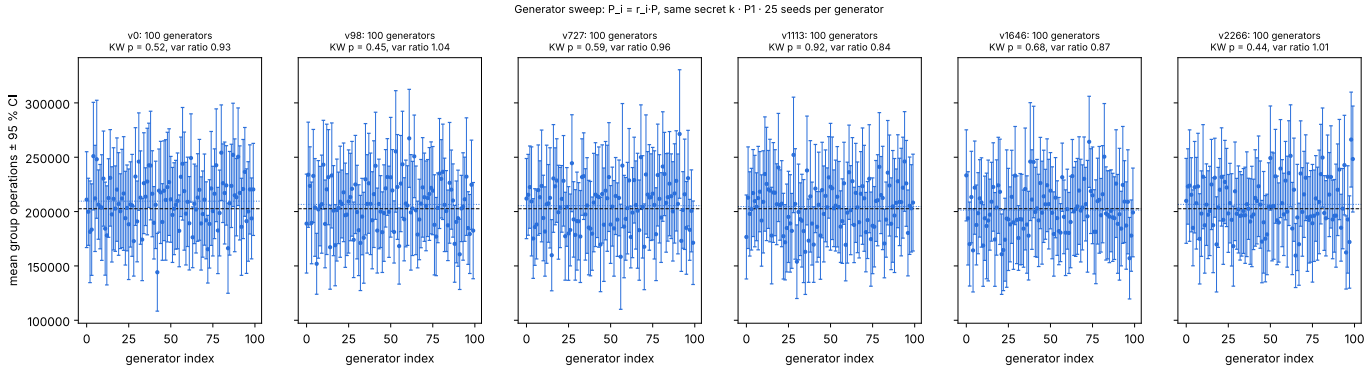


Figure 9: P1 generator sweep per curve

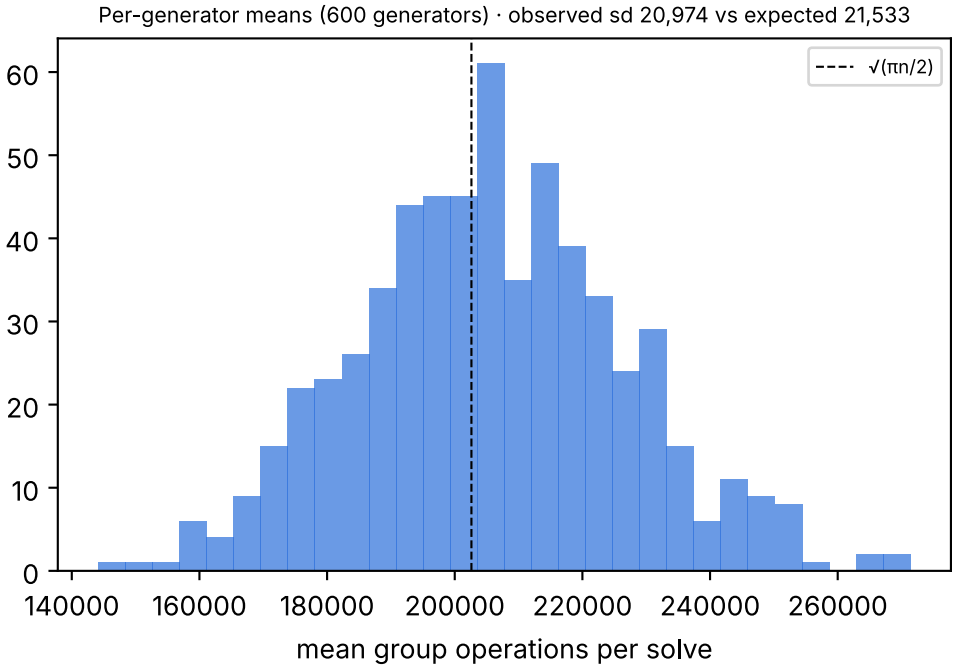


Figure 10: P1 distribution of generator means

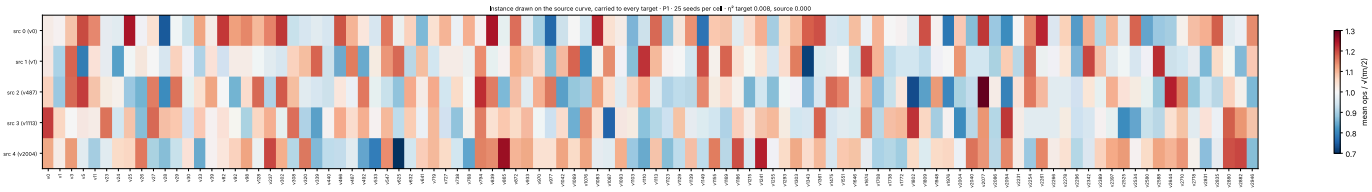


Figure 11: P1 source × target matrix

P0

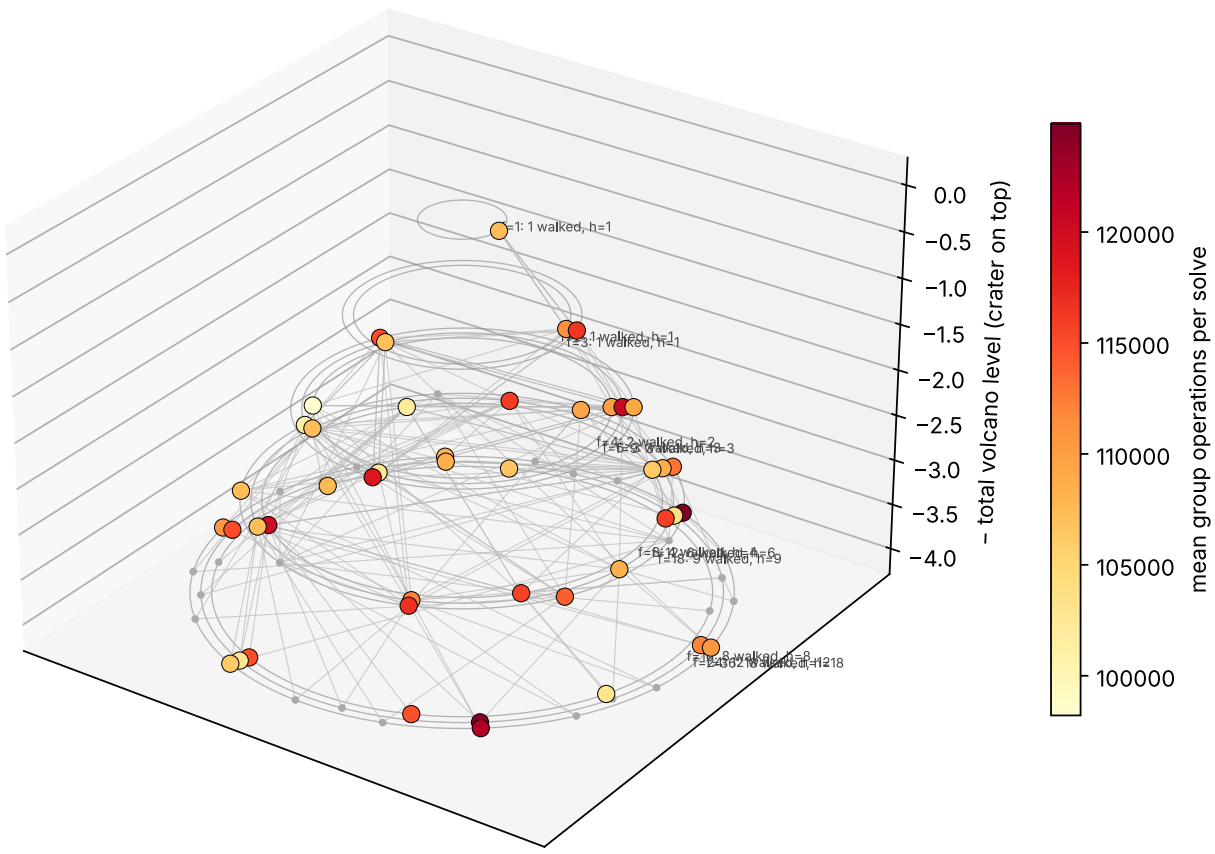


Figure 12: P0 3D class heat map, mean group operations per solve

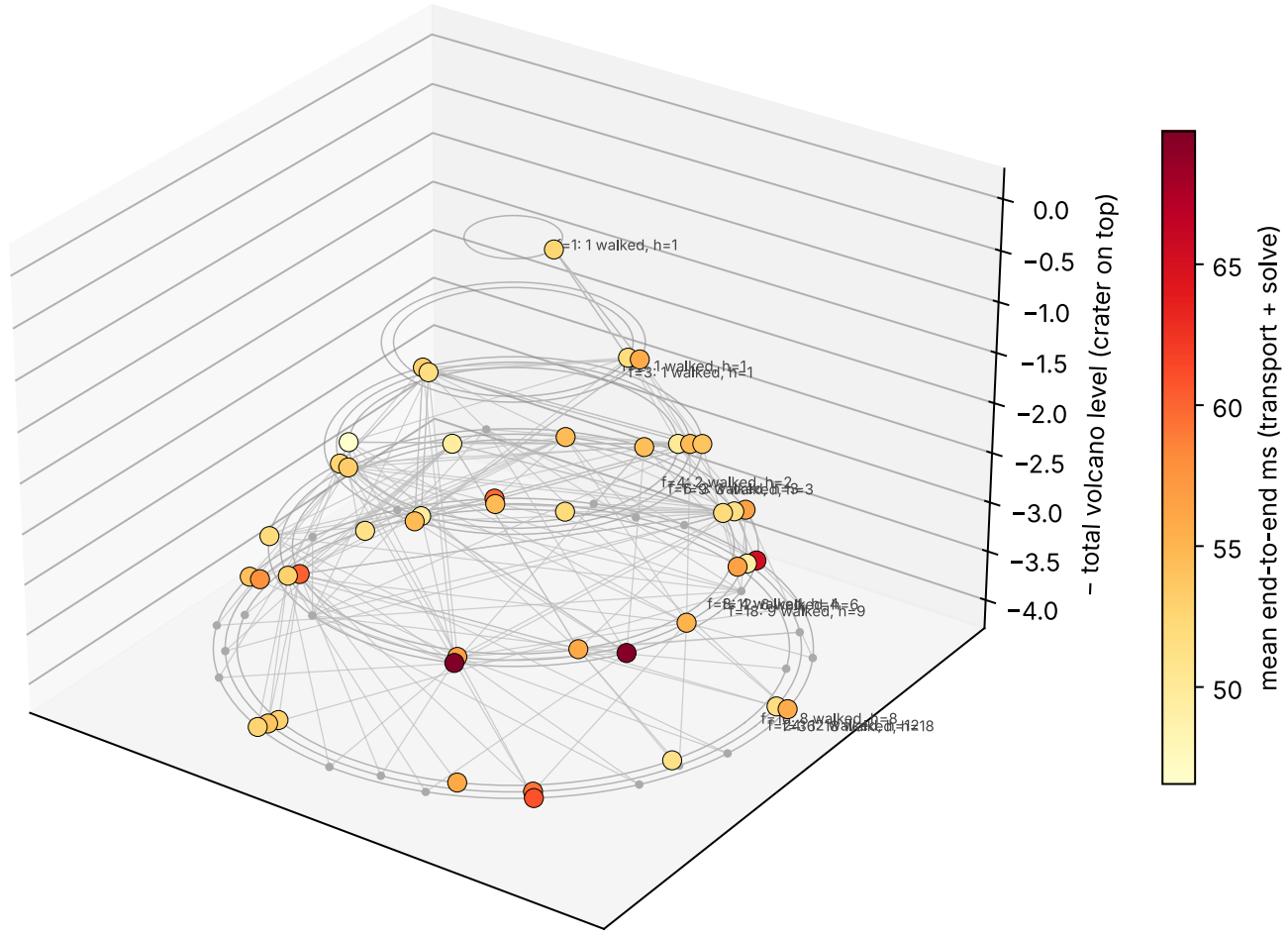


Figure 13: P0 3D class heat map, mean end-to-end ms (transport + solve)

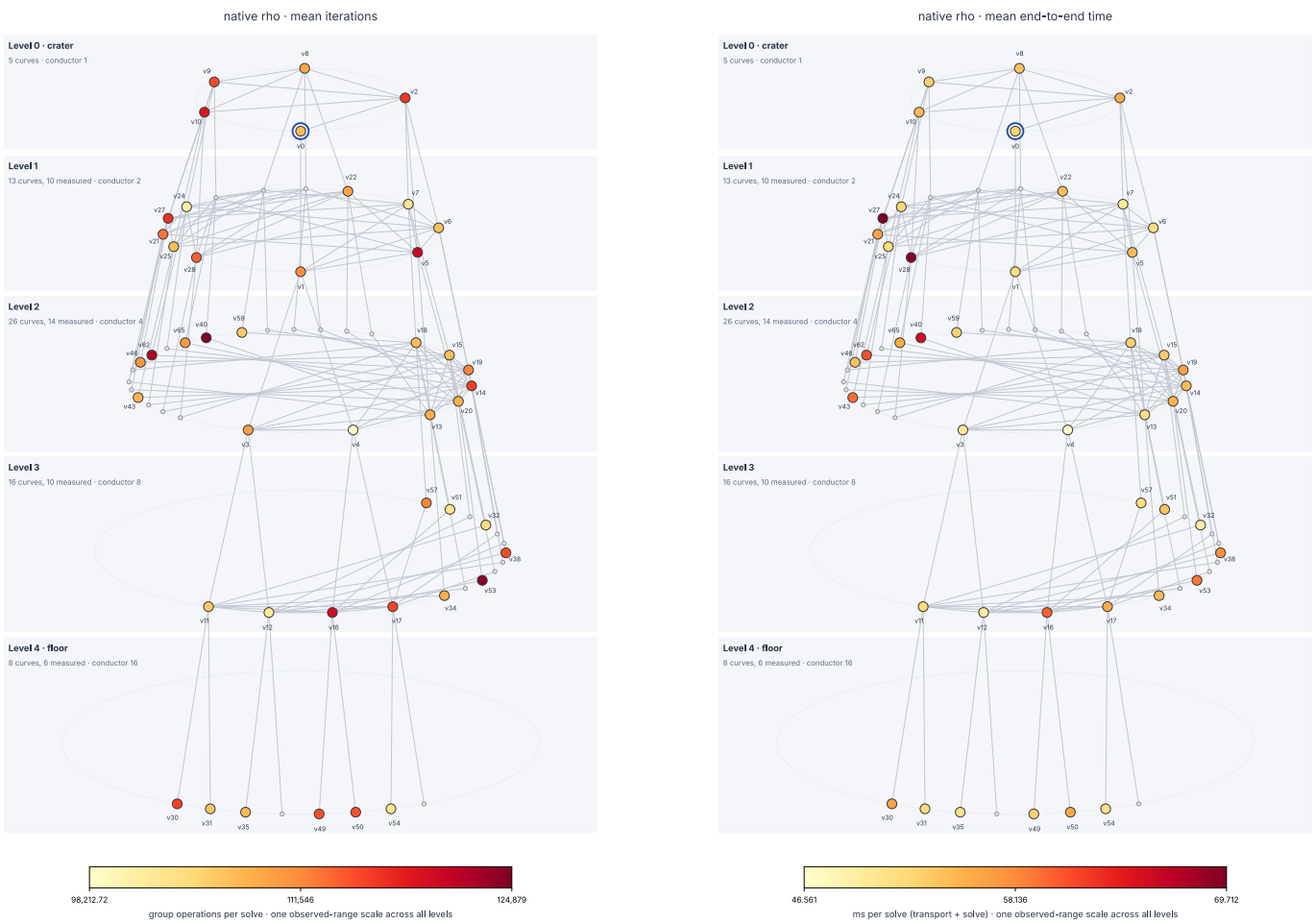


Figure 14: P0 volcano-ell2: standard volcano diagram

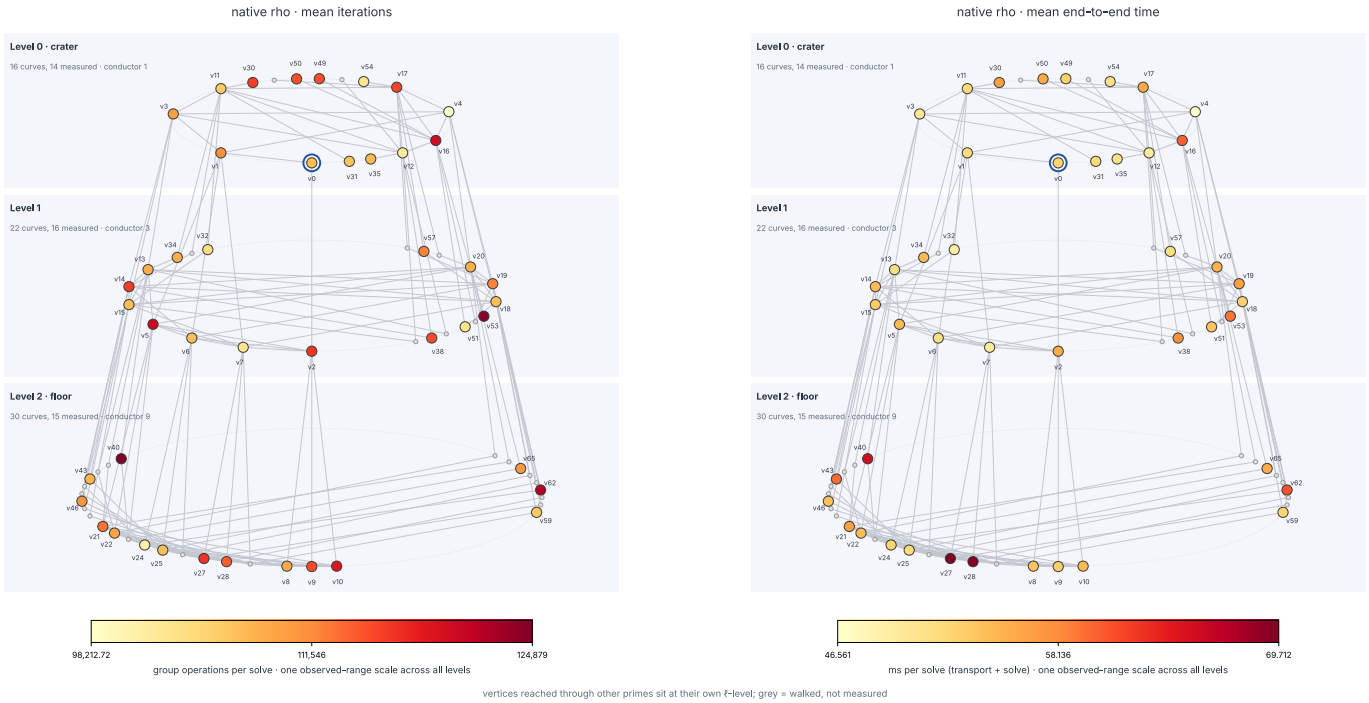


Figure 15: P0 volcano-ell3: standard volcano diagram

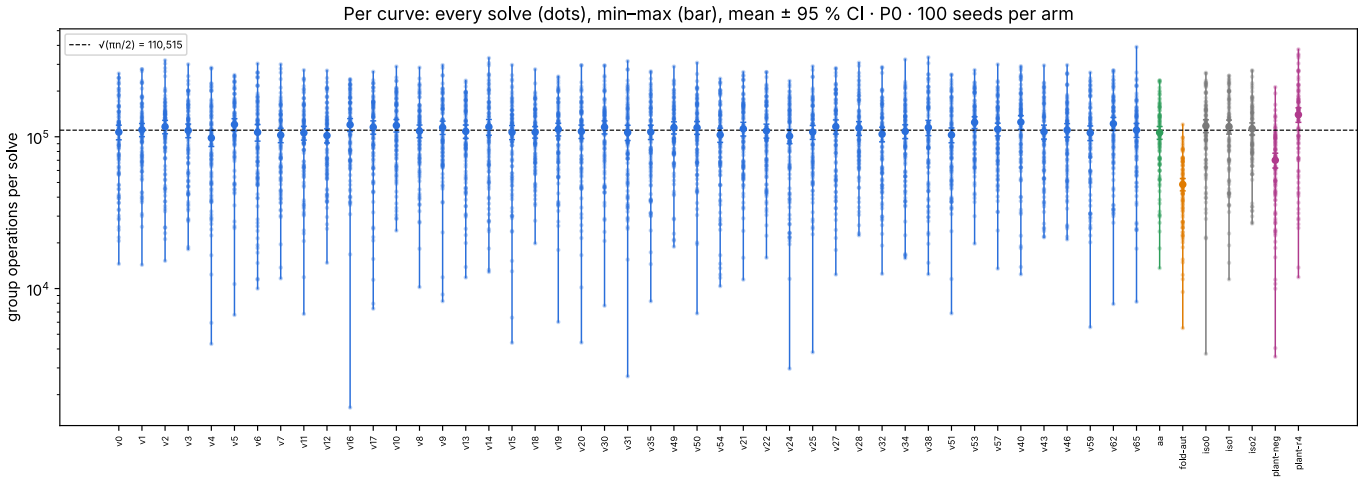


Figure 16: P0 every solve, min–max, mean ± CI per arm

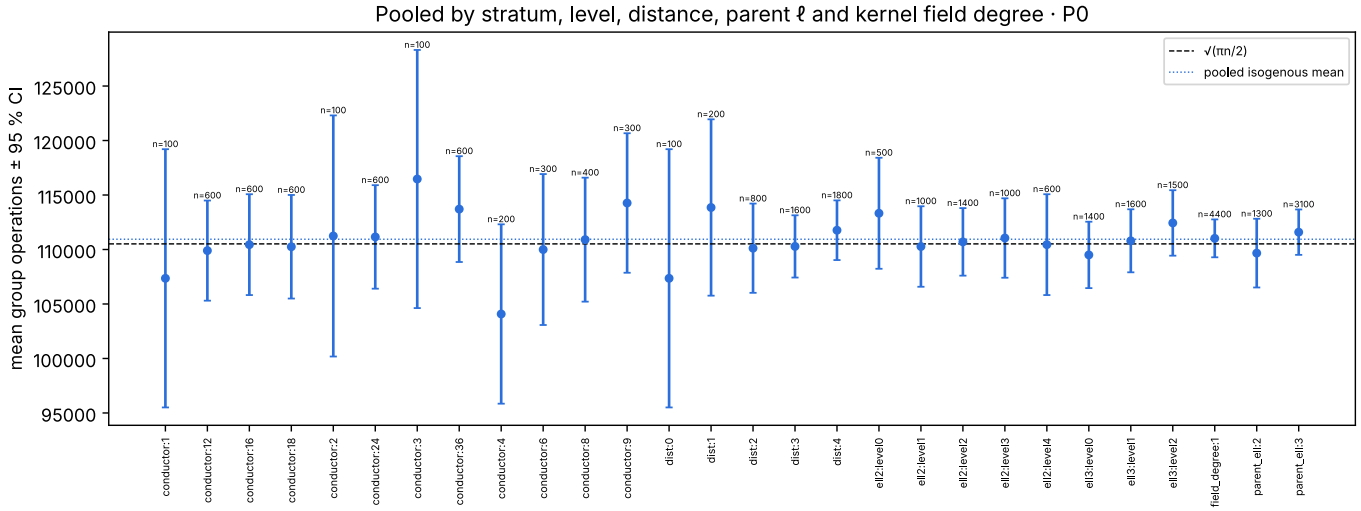


Figure 17: P0 pooled groups

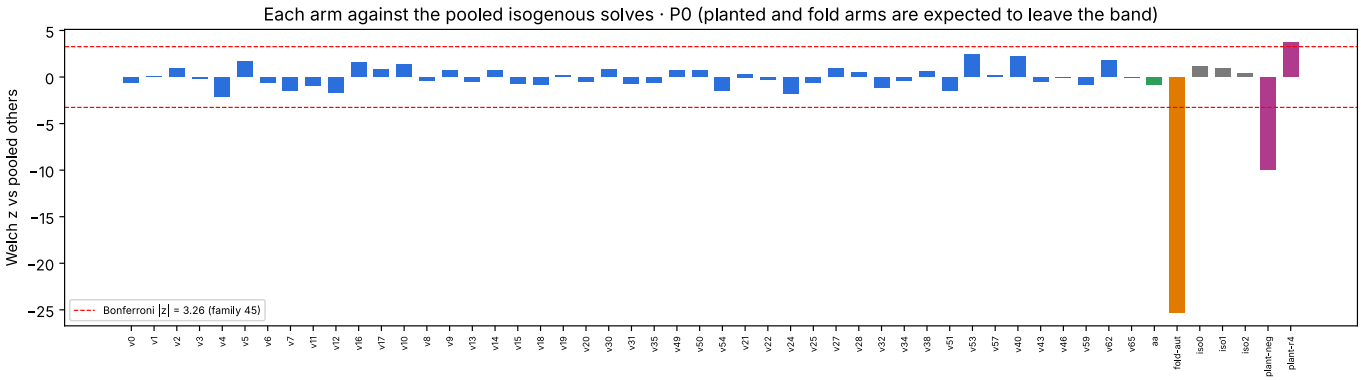


Figure 18: P0 each arm against the pooled isogenous solves

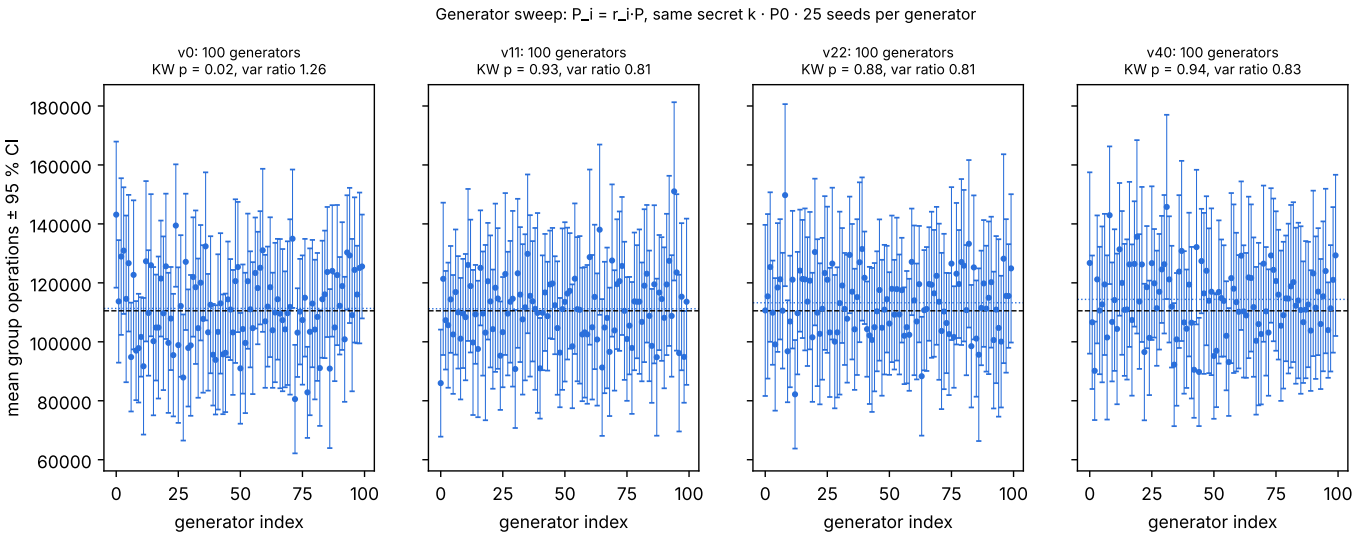


Figure 19: P0 generator sweep per curve

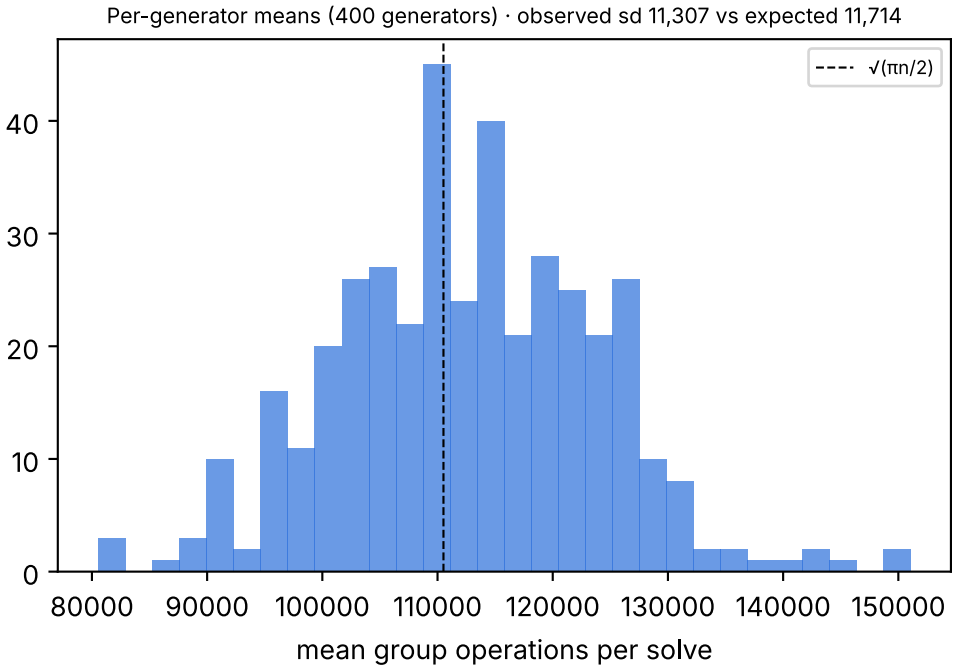


Figure 20: P0 distribution of generator means

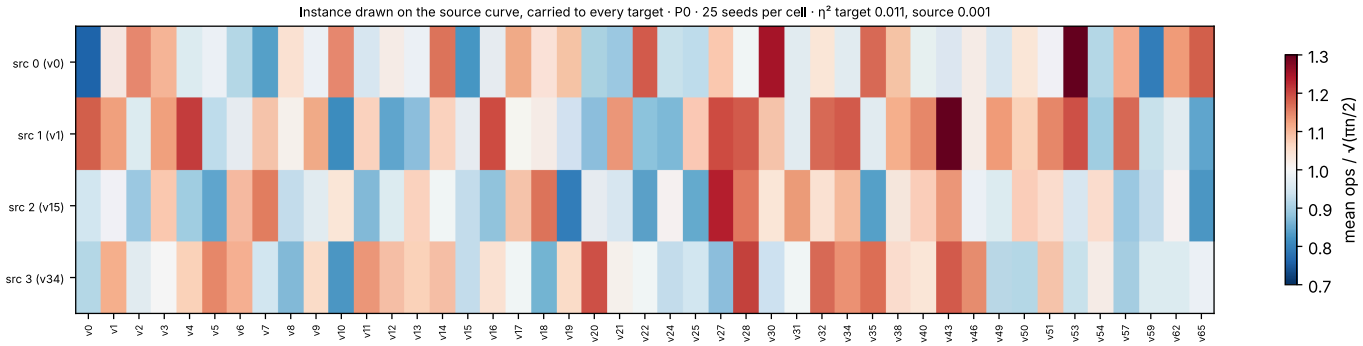


Figure 21: P0 source × target matrix

3D isogeny-class heat map · F_2^37 · mean group operations per solve
B1 · rings = endomorphism-ring strata (conductor f of End), 40 measured of 177 walked curves · expected √(πn/2) = 328,549

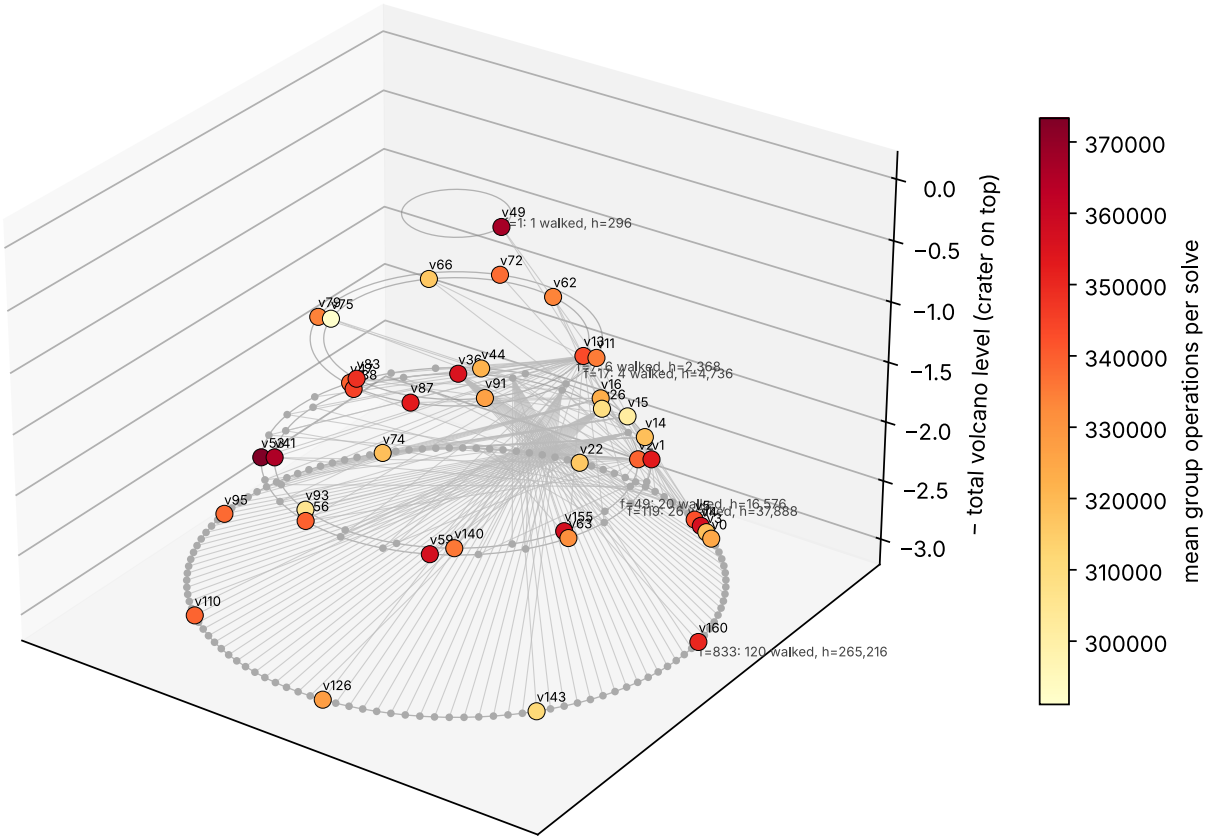


Figure 22: B1 3D class heat map, mean group operations per solve

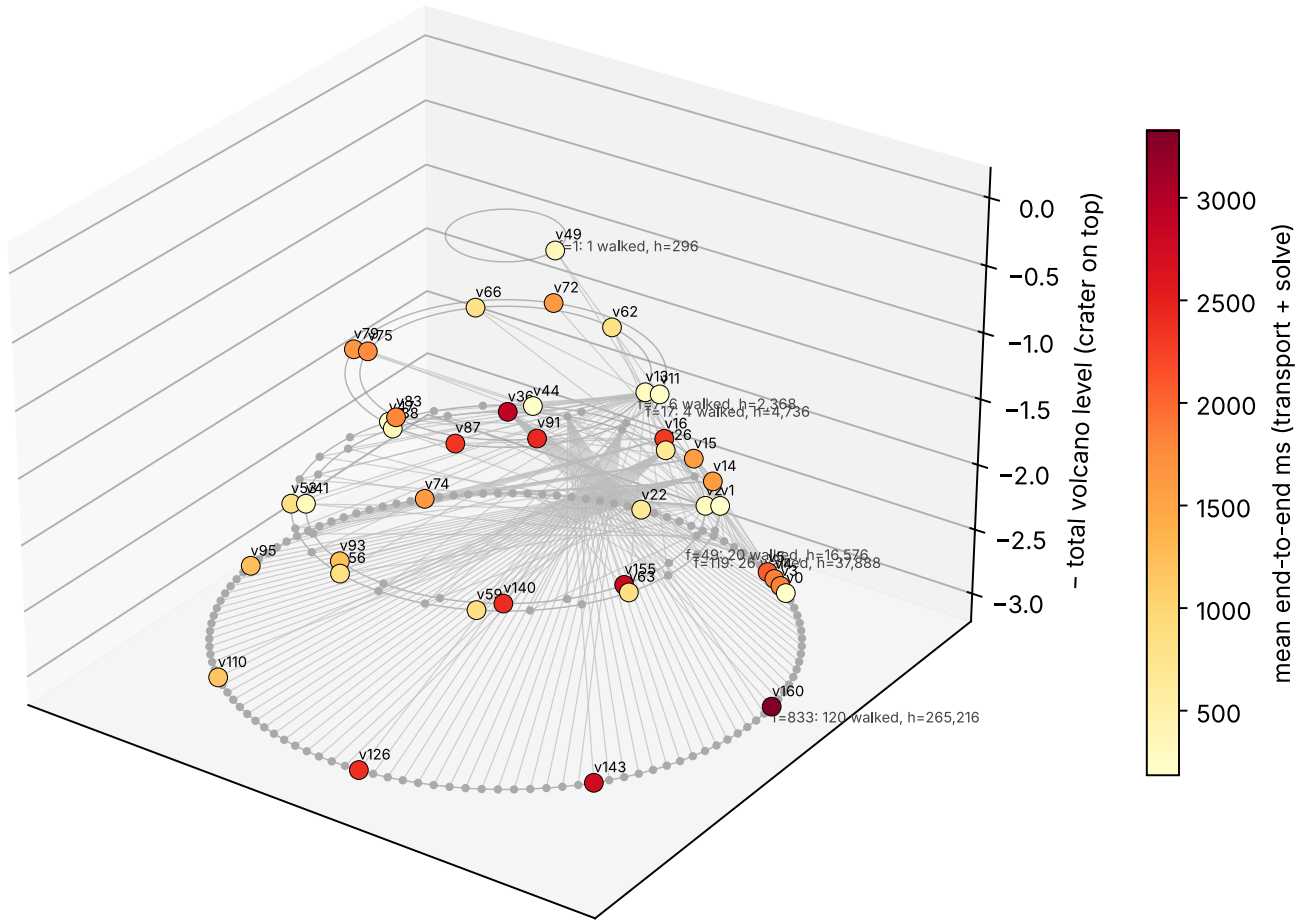


Figure 23: B1 3D class heat map, mean end-to-end ms (transport + solve)

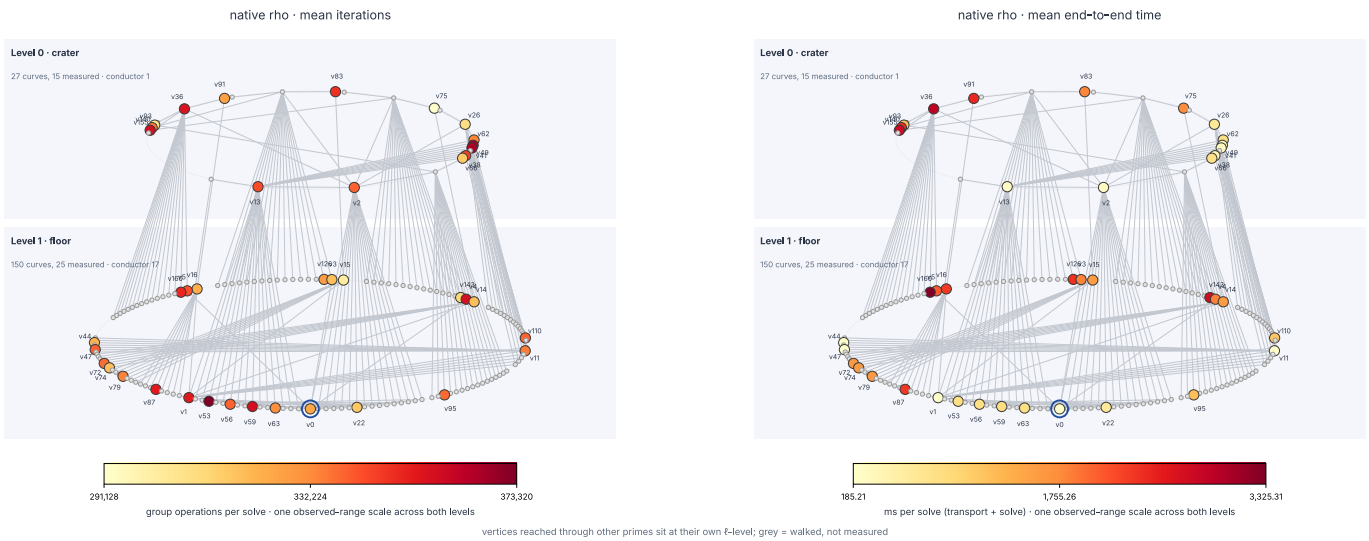


Figure 24: B1 volcano-ell17: standard volcano diagram

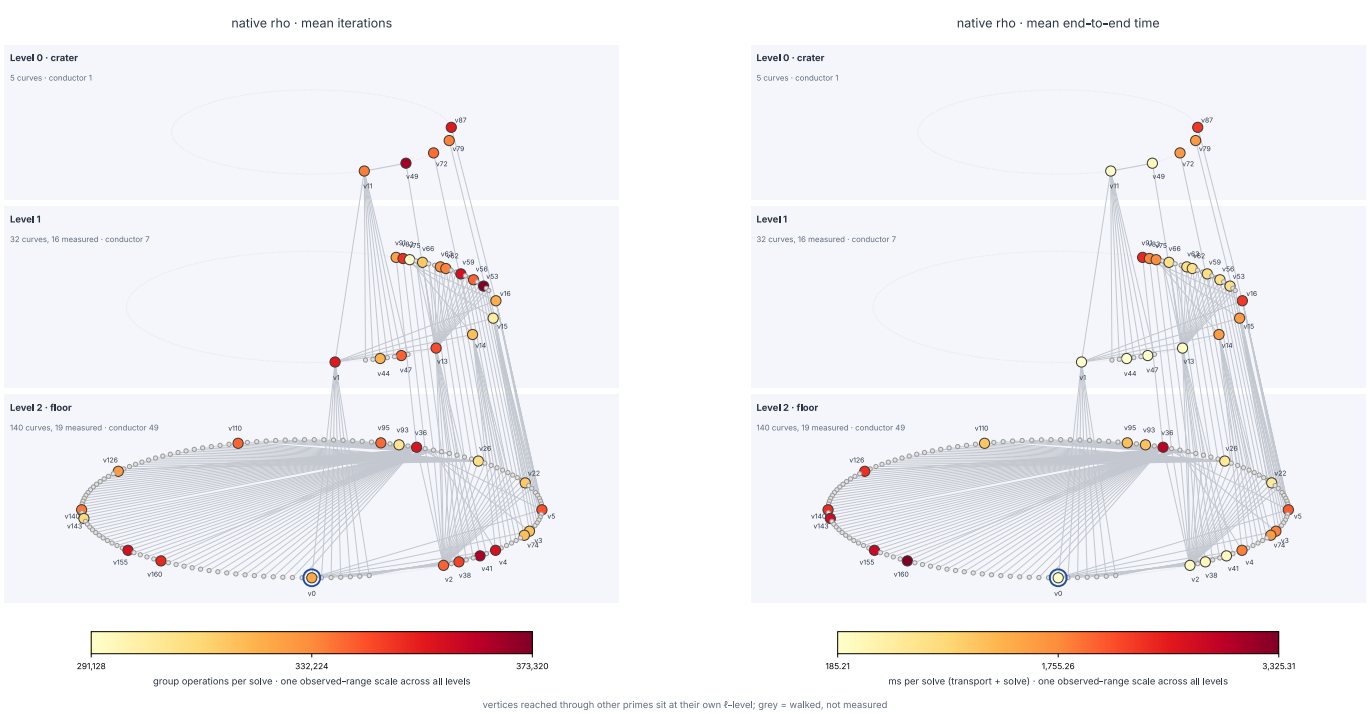


Figure 25: B1 volcano-ell7: standard volcano diagram

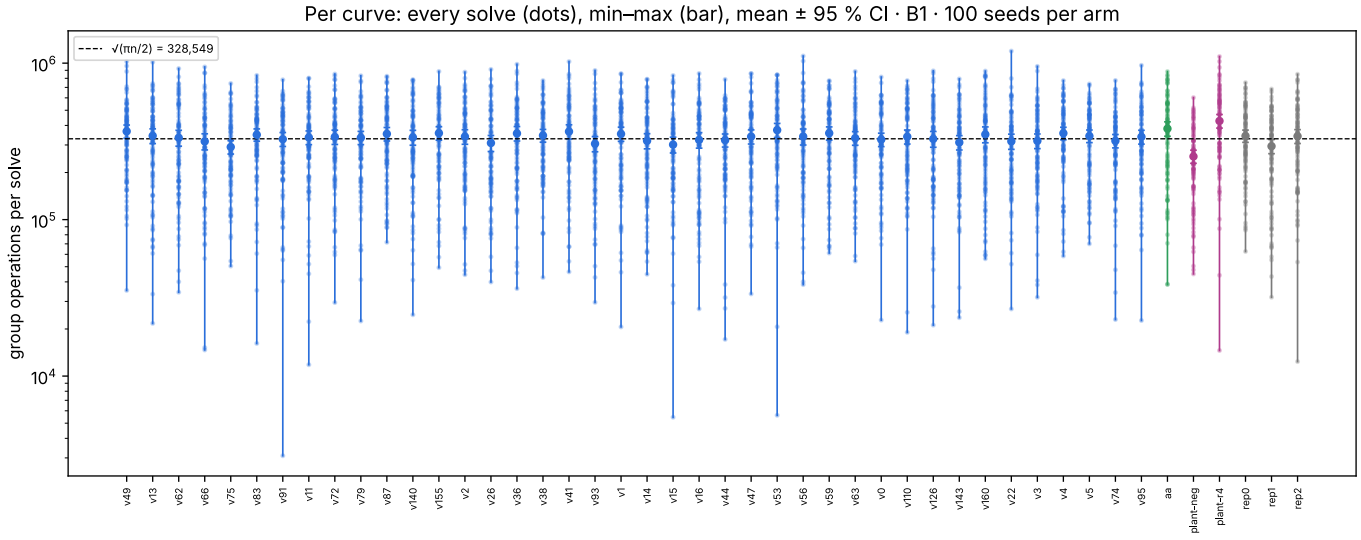


Figure 26: B1 every solve, min–max, mean ± CI per arm

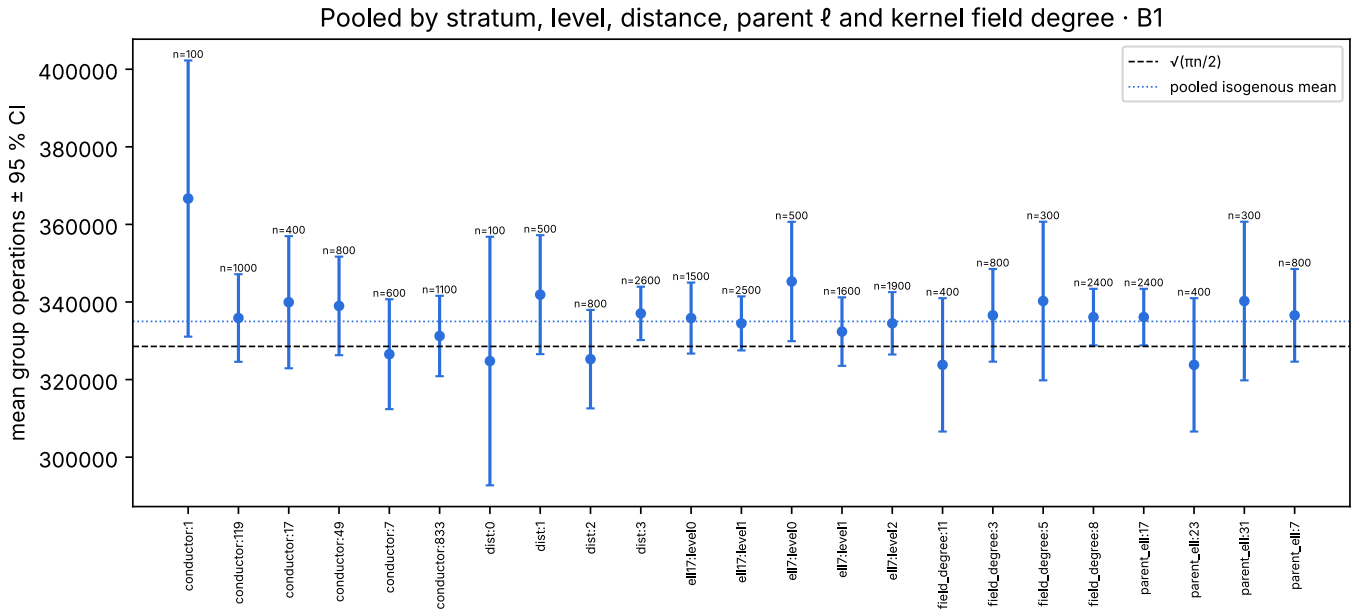


Figure 27: B1 pooled groups

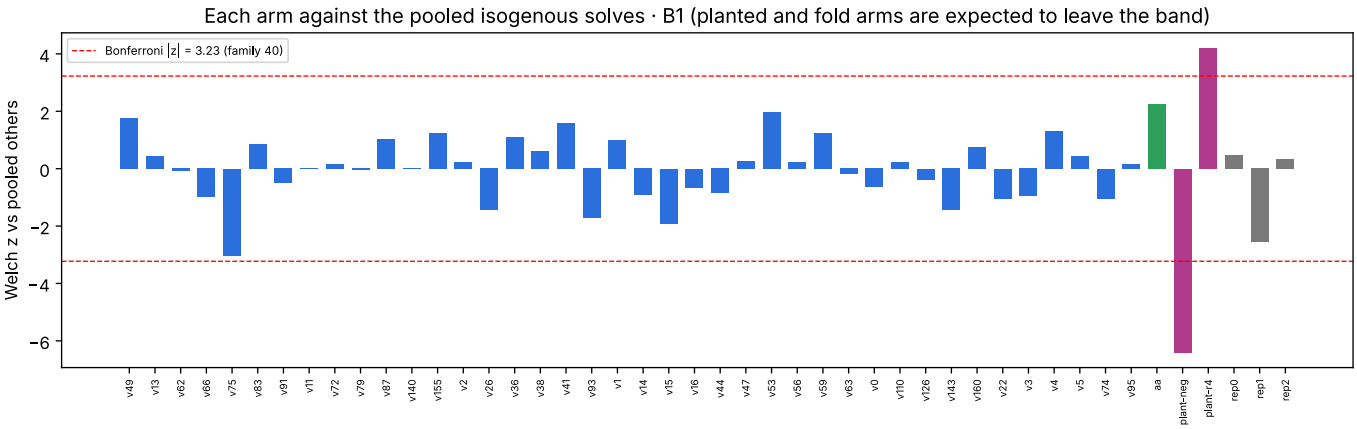


Figure 28: B1 each arm against the pooled isogenous solves

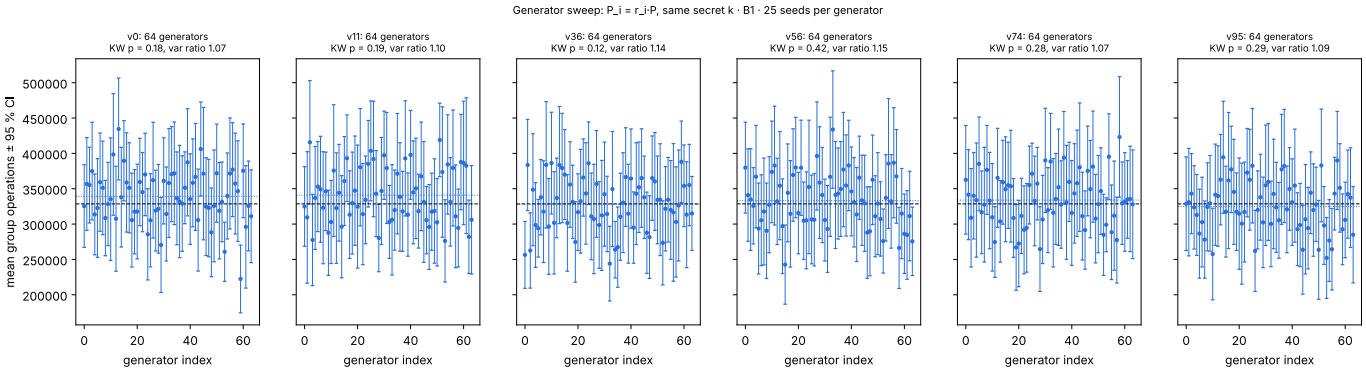


Figure 29: B1 generator sweep per curve

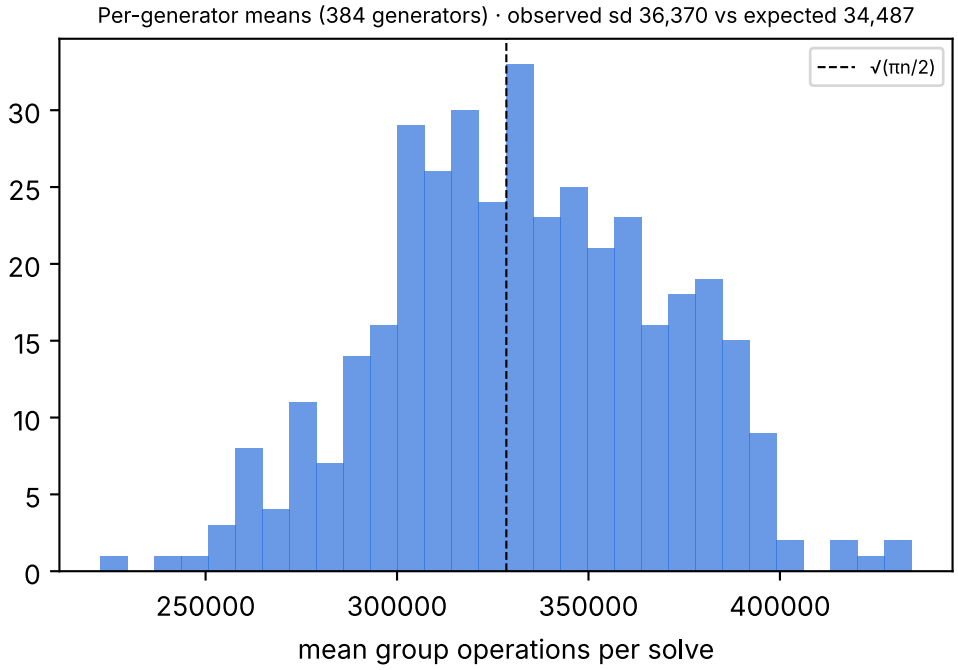


Figure 30: B1 distribution of generator means

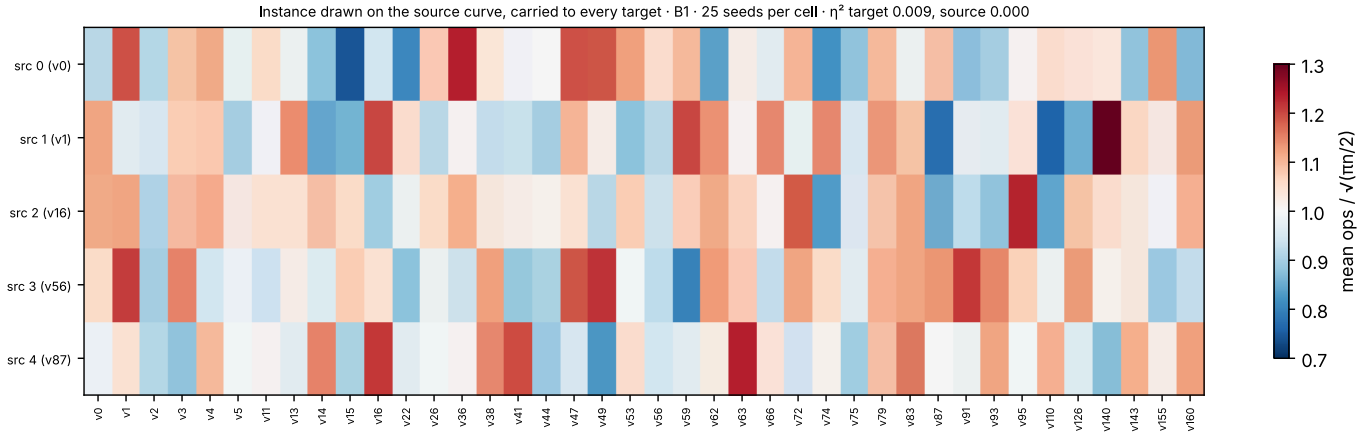


Figure 31: B1 source × target matrix

K

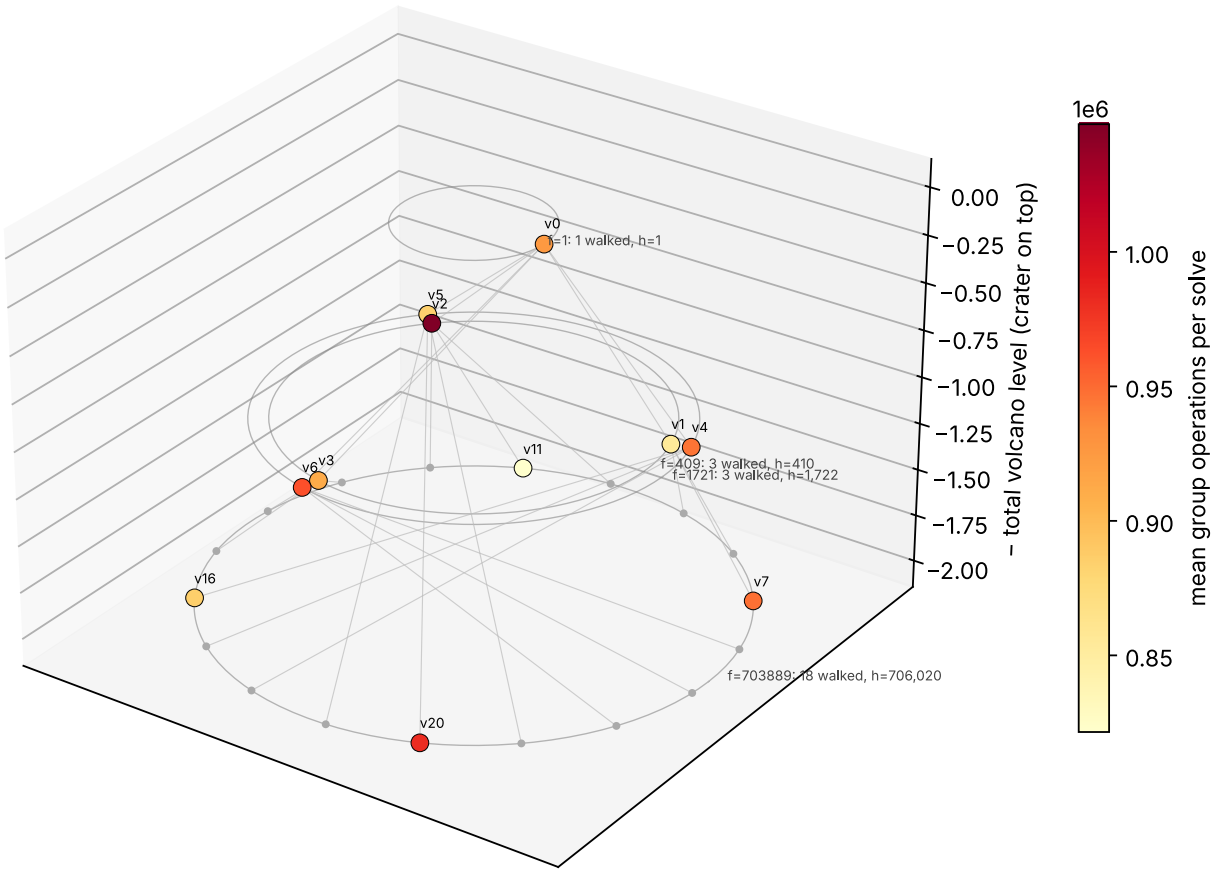


Figure 32: K 3D class heat map, mean group operations per solve

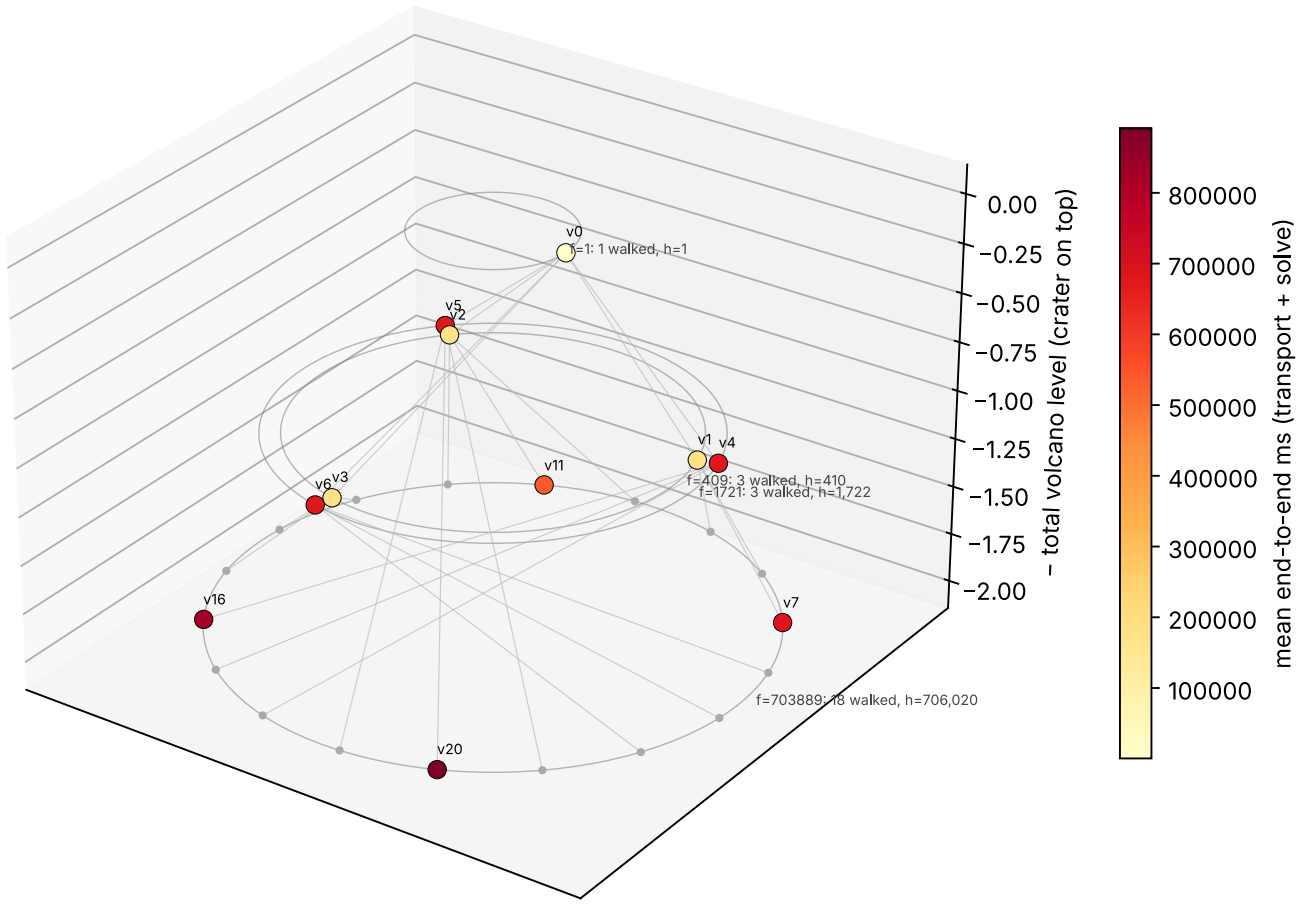


Figure 33: K 3D class heat map, mean end-to-end ms (transport + solve)

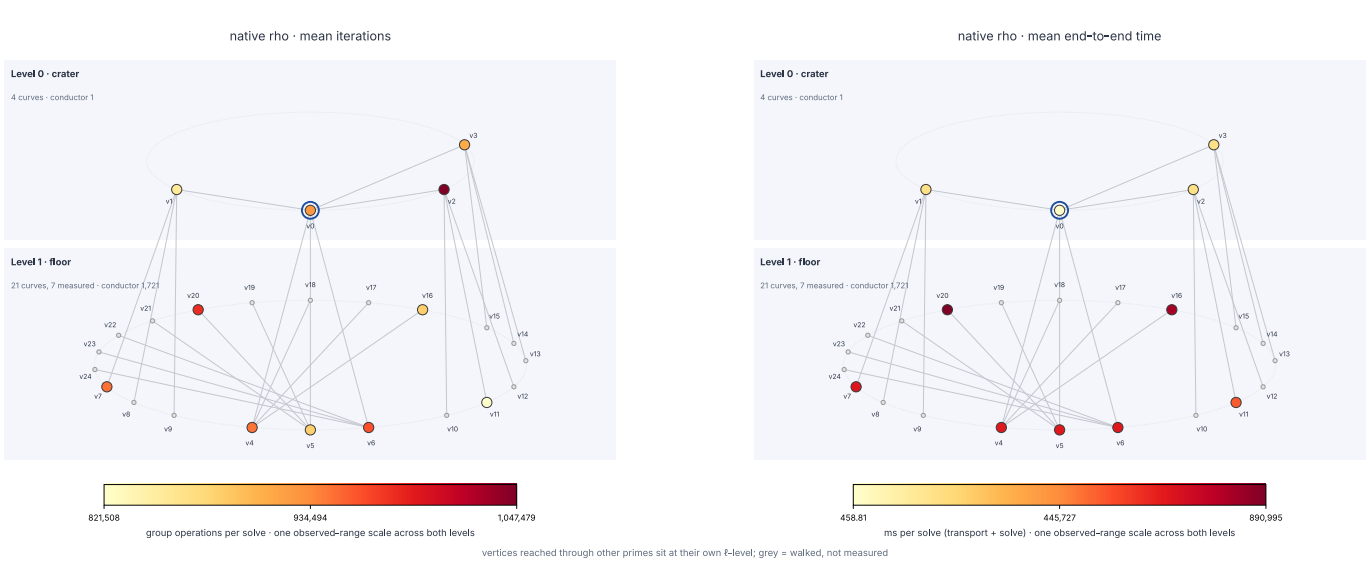


Figure 34: K volcano-ell1721: standard volcano diagram

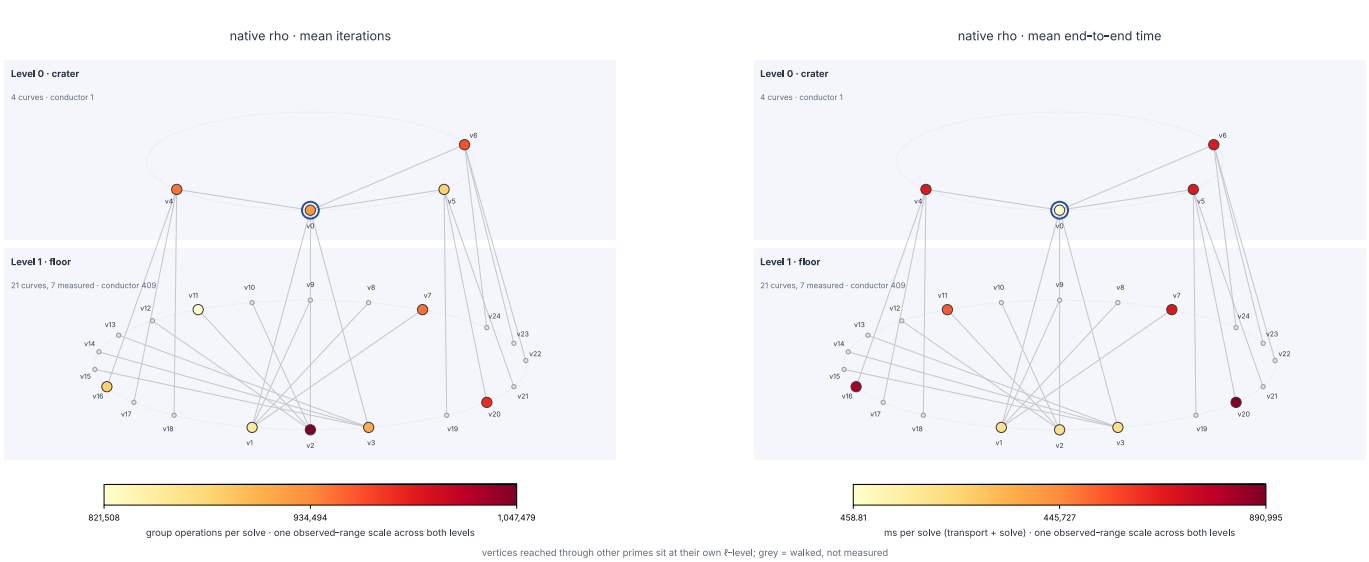


Figure 35: K volcano-ell409: standard volcano diagram

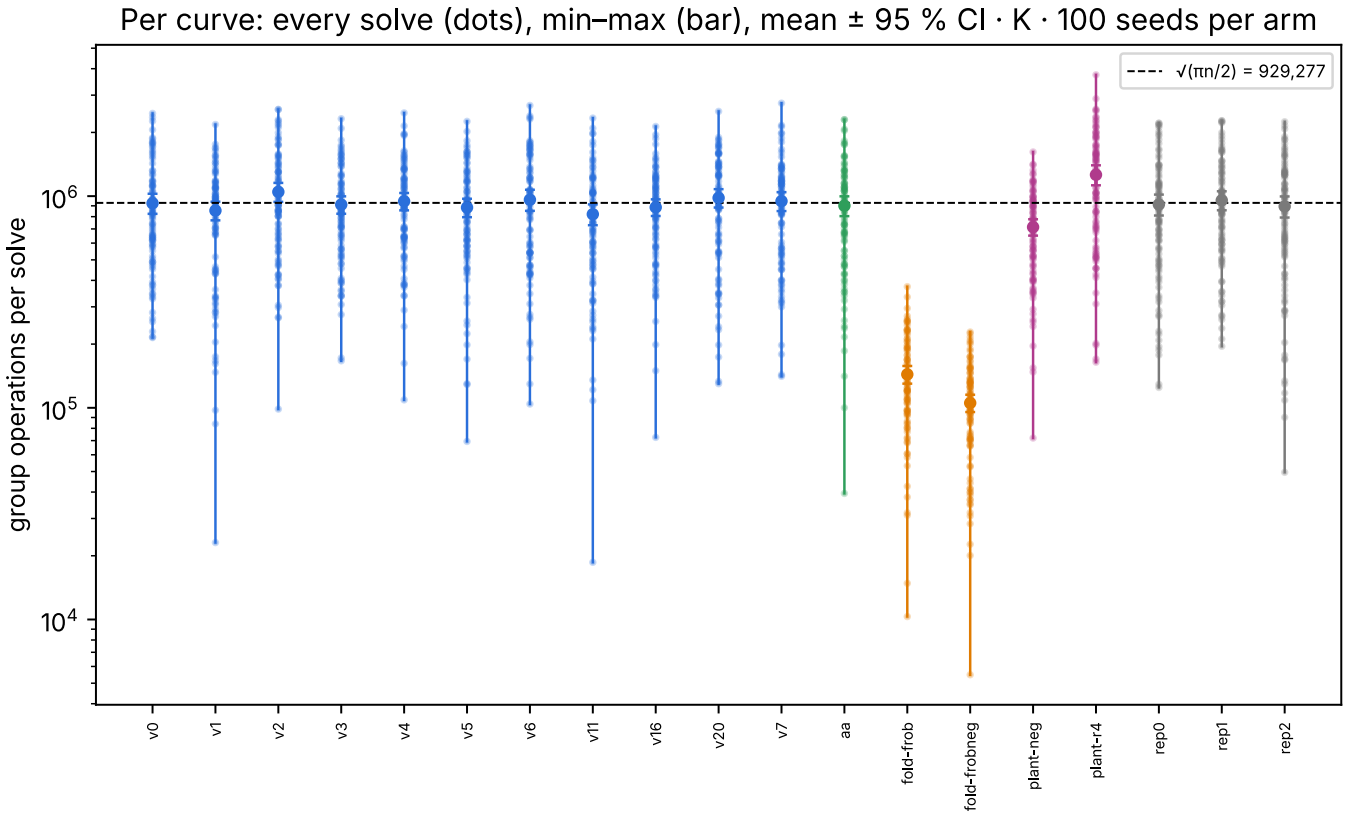


Figure 36: K every solve, min–max, mean ± CI per arm

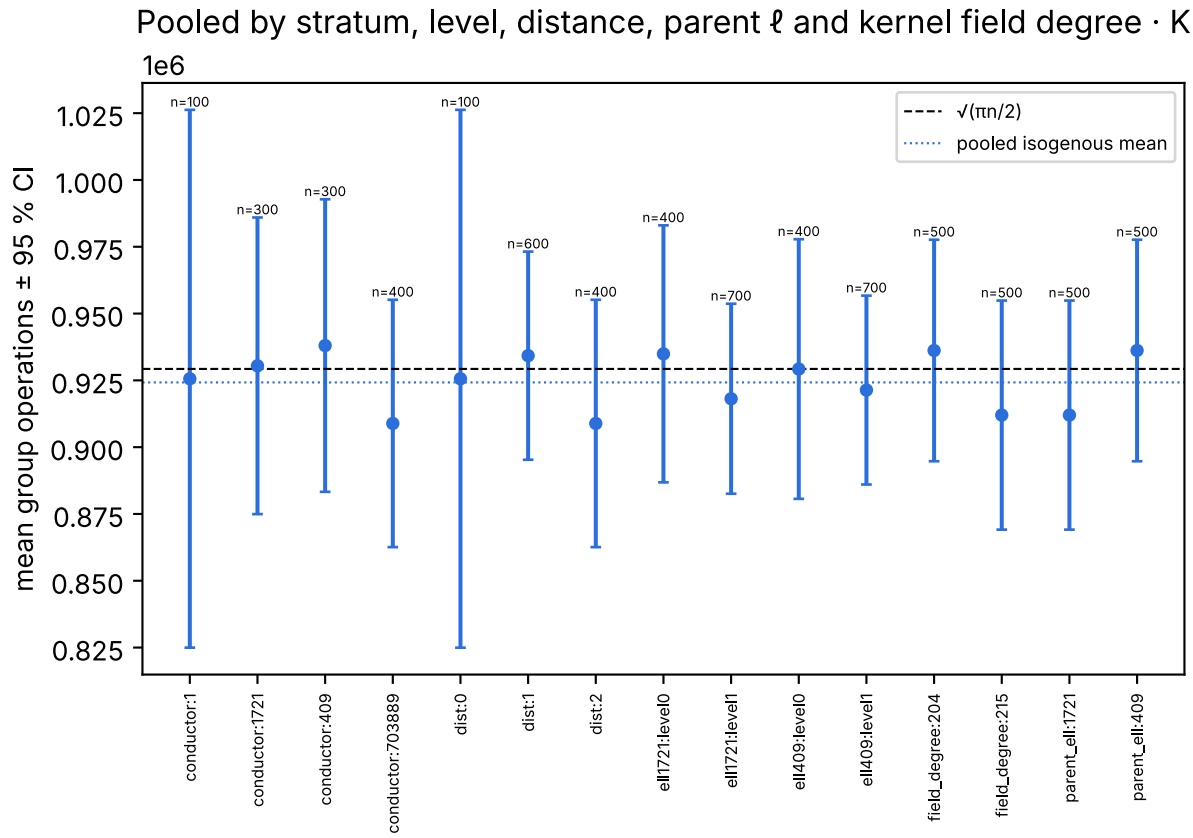


Figure 37: K pooled groups

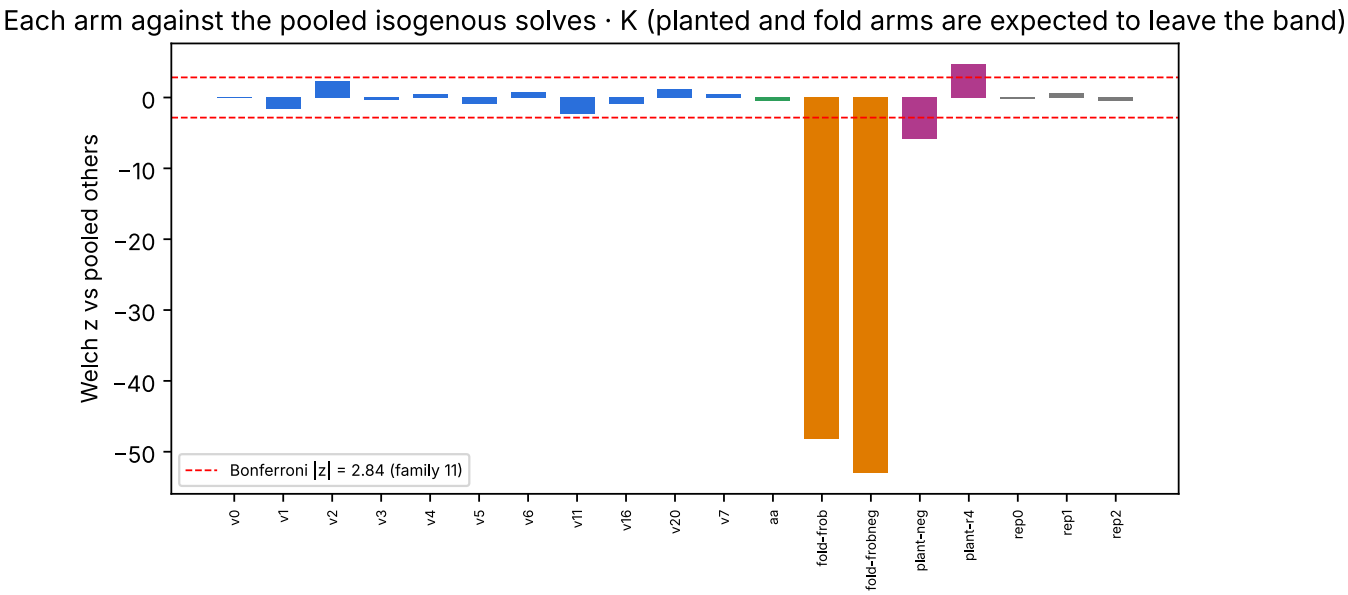


Figure 38: K each arm against the pooled isogenous solves

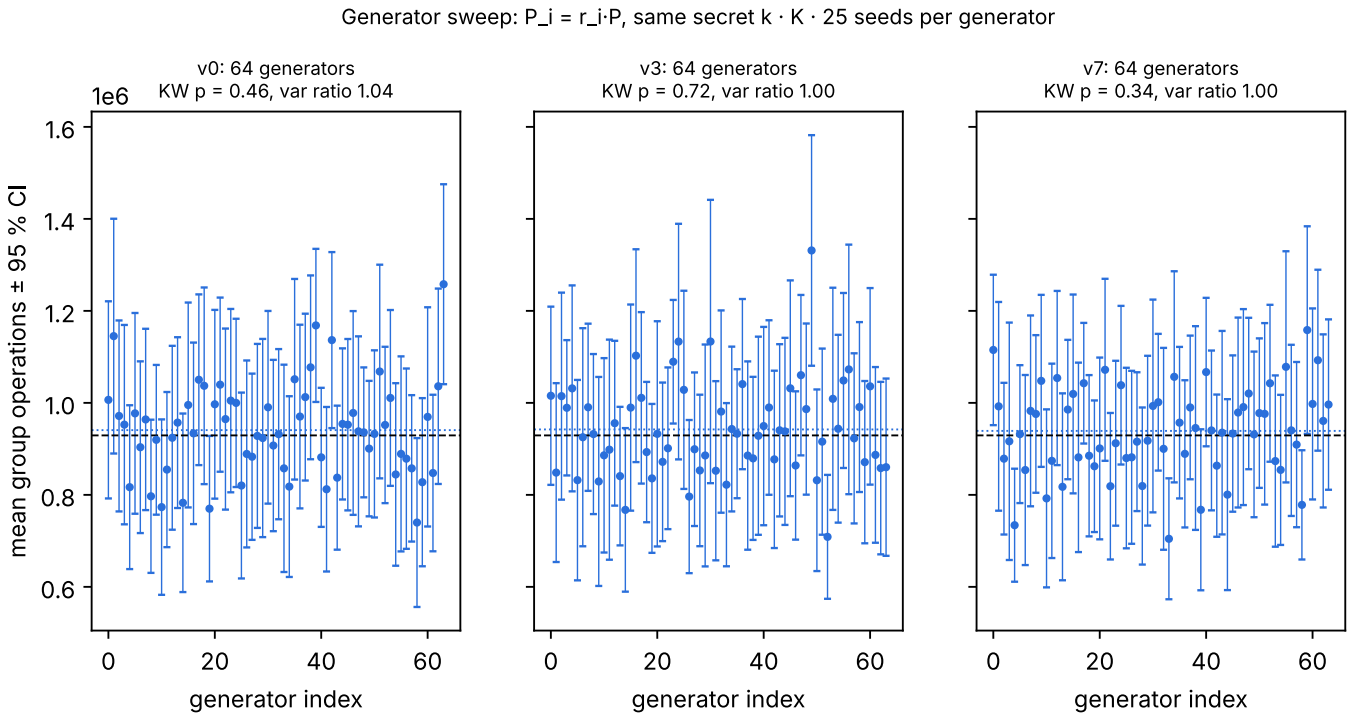


Figure 39: K generator sweep per curve

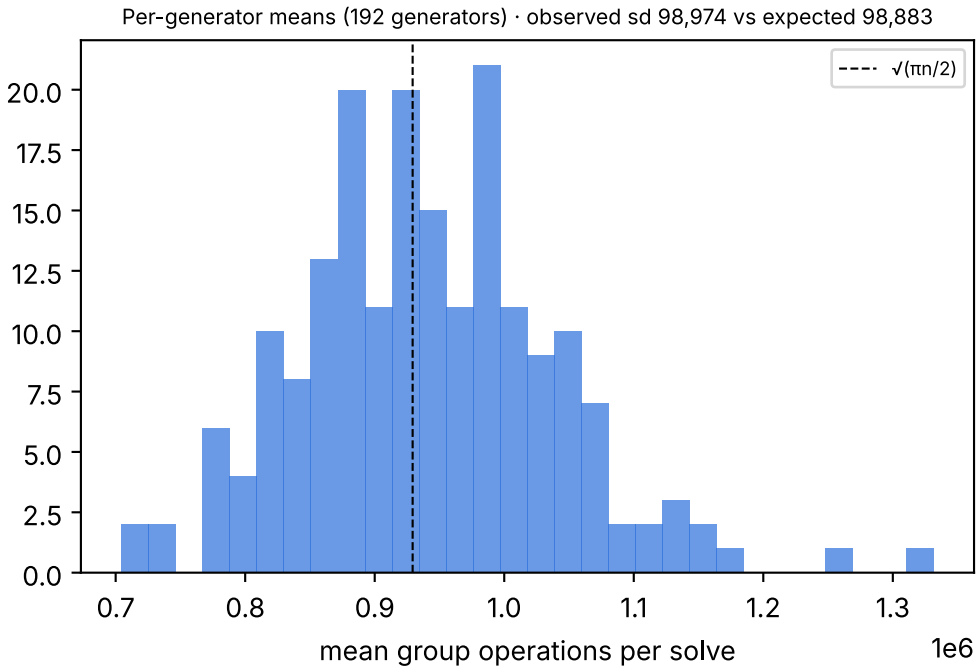


Figure 40: K distribution of generator means

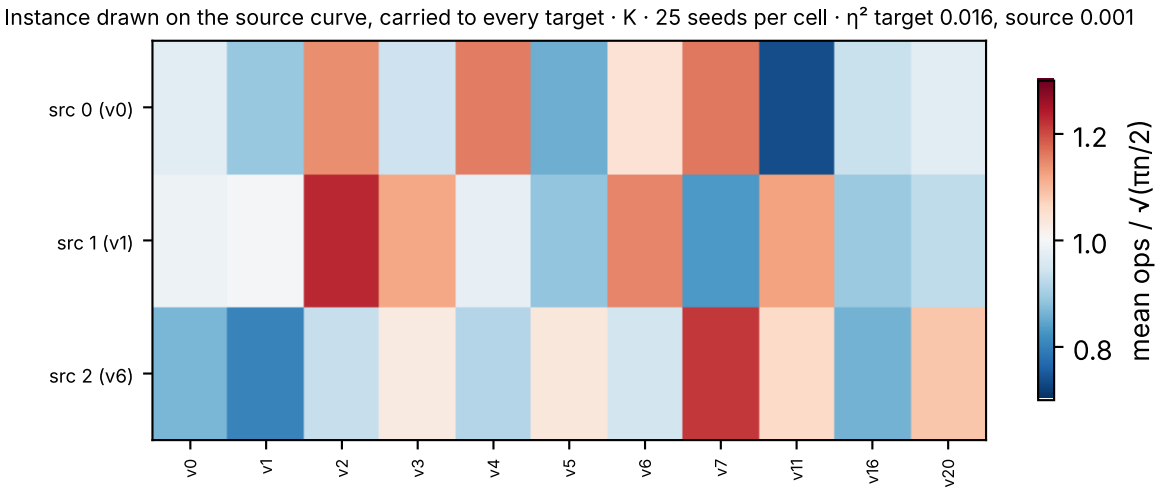


Figure 41: K source × target matrix

Reading (verdicts by the protocol's rules; measured unless marked

derived) Verdict wording follows PROTOCOL.md §4. "Within the control distribution" means no arm or group left the Bonferroni band and the across-arm tests did not detect heterogeneity; it is not "no effect exists" below the stated resolution (per curve about 18 %, per stratum about 6.5 %, class-wide about 2 % at these sample sizes).

P1 (prime field, class of 347,004 curves, 104 measured over 12

strata)

- **R1, per curve:** every one of the 104 curves is within the control distribution (largest |z| 3.03 against the threshold 3.49; none flagged). Pooled mean 1.020 × √(πn/2) with the 32-adding walk's expected 1.016.
- **R2, across curves:** heterogeneity not detected (KW p = 0.089, ANOVA p = 0.101, variance ratio 1.18: the spread of the 104 curve means is 18 % larger than the pooled single-solve spread predicts, inside what 104 means of 100 skewed solves produce). The A/A replica (z = 0.70) and the three isomorphic models (z = 2.58, 1.00, 0.54) are inside the band; the first isomorphic model is the most extreme arm of the whole class mode, which is the right calibration: a change of coordinates reaches |z| = 2.6 at 100 seeds.
- **R3, by stratum / level / distance / degree / kernel field:** all 12 endomorphism-ring strata (conductors 1 … 315) are within ±5 % of the reference with |z| ≤ 1.6; the three distances from the root, every parent degree ℓ ∈ {3, 5, 7, 13, 19, 31, 41, 43, 47} and every kernel field degree d ∈ {1, 2, 3, 4, 5, 9, 15, 21, 23} likewise (|z| ≤ 1.6). The curves reached through the hard isogenies (ℓ = 31 … 61, kernel abscissae in F_{p^15} … F_{p^23}) cost the same as those reached through ℓ = 3.
- **R4, positive controls:** `plant-neg` 0.725 [0.651, 0.804] (expected 0.707, z = −7.5): detected. `plant-r4` 1.298 [1.172, 1.440] (Teske's heuristic 1.137, z = +4.0): detected, and larger than the heuristic (the 4-adding walk is further from random than the heuristic assumes; the dry runs and P0 show the same). The detector sees a 28 % effect at |z| ≈ 7 and a 14–30 % one at |z| ≈ 4 on a single arm.
- **R5, generators:** 6 curves × 100 generators × 25 seeds, 15,000 solves, all correct. On every curve the spread of the 100 generator means matches the null (sd 19,852–21,863 against 21,275–21,873 expected; variance ratios 0.84–1.04; KW p 0.44–0.92); no generator flagged (family 600, largest |z| 3.76 against 3.93); Spearman correlations of the generator mean with r, bits(r), popcount(r) and x(P_i) are all |ρ| ≤ 0.12. Which generator of ⟨P⟩ carries the instance does not matter.
- **R6, shift:** 5 sources × 104 targets × 25 seeds, 13,000 solves, all correct. η^2(target) = 0.0085, η^2(source) = 0.0002; KW by target p = 0.19, by source p = 0.37; the five sources' pooled means are 1.007–1.029 × reference (|z| ≤ 1.3). Two of the 520 cells sit beyond the Bonferroni band on the low side (s1⟶v1343 at 0.72 and s4⟶v625 at 0.70, Welch z ≈ −4.5, Mann–Whitney z −2.9 and −3.1): both targets are unremarkable in the other four rows and in the class mode (v625 1.013, v1343 1.036), so this is the low tail of 520 small cells, not a target or source effect. Where the instance was drawn does not matter.

P0 (j = 0 crater, class of 38,472 curves, 45 measured over 12

strata)

- **R1–R3:** all 45 curves within the control distribution (largest |z| 2.42 against 3.26); heterogeneity not detected (KW p = 0.16, variance ratio 1.13); every stratum (conductors 1 … 36), every level of the 2-volcano (0 … 4) and of the 3-volcano (0 … 2) within ±6 % with |z| ≤ 1.7. Controls inside the band (iso 1.17, 0.87, 0.40; A/A −0.83).
- **R4:** `plant-neg` 0.634 [0.567, 0.709] (z = −10); `plant-r4` 1.265 [1.136, 1.411] (z = +3.7); **`fold-aut` 0.439 [0.401, 0.481] against the expected 1/√6 = 0.408** (z = −25): the order-6 automorphism group of the crater curve, folded into the walk, cuts the operation count by 2.28×. That fold exists only on the j = 0 curve: every other curve of the class has Aut = ±1, and the unfolded walk costs the same on all 45. The isogeny carries the instance away from the one curve where the cheap endomorphism lives; it does not create one.

B1 (binary field F_2^37, class of 327,080 curves, 40 measured over

6 strata)

- **R1–R3:** all 40 curves within the control distribution (largest |z| 3.02 against 3.23); heterogeneity not detected (KW p = 0.16, variance ratio 1.11); the six endomorphism-ring strata (conductors 1, 7, 17, 49, 119, 833: the crater and every order down to Z[π]) within ±12 % with |z| ≤ 1.8 (the crater is one curve); every level of the 7-volcano (0, 1,

  2) and of the 17-volcano (0, 1) within ±5 %; curves reached through the hard 23- and 31-isogenies (kernel abscissae in F_{2^{37·11}} and F_{2^{37·5}}, 1.3–2.4 s each to find) cost the same as those reached through ℓ = 7 (ratios 0.986 and 1.036, |z| ≤ 1.3). The three representation controls (the same curve in three other polynomial bases of the field) sit at 1.04, 0.90, 1.04 (|z| ≤ 2.6) and the A/A replica at 1.16 (z = 2.2): a change of field basis or of seeds moves a 100-seed mean as much as any curve does.
- **R4:** `plant-neg` 0.771 [0.702, 0.847] (z = −6.4); `plant-r4` 1.300 [1.177, 1.435] (z = +4.2). Detected.
- **R5:** 6 curves × 64 generators × 25 seeds, 9,600 solves, all correct. Per curve the spread of generator means is 1.07–1.15× the null (KW p 0.12–0.42); one generator of 384 (v0, generator 59, r = 51151694715) is beyond the Bonferroni band on the low side (0.68 [0.54, 0.83], Welch z −4.5, Mann–Whitney −3.3), the

all-arm KW p = 0.039. Its curve's other 63 generators and the same r-features show nothing (|ρ| ≤ 0.27); recorded as the one flagged cell of this family, consistent with the low tail of 384 means of 25 skewed solves, not reproduced elsewhere.

- **R6:** 5 sources × 40 targets × 25 seeds, 5,000 solves, all correct. η^2(target) = 0.0089, η^2(source) = 0.0003; KW by target p = 0.15, by source p = 0.56; no cell flagged (largest |z| 3.17 against 3.66).

K (Koblitz E_0 over F_2^41, class of 708,153 curves, 11 measured
over all 4 strata)

- **The class.** `D = 703889² · (−7)`, `703889 = 409 · 1721`, `h(−7) = 1`: the Koblitz curve is alone on the crater of both volcanoes, every horizontal edge is a loop, and the class is the crater plus `h(409²·(−7)) = 410`, `h(1721²·(−7)) = 1,722` and `h(703889²·(−7)) = 706,020` curves (the last counted directly as well as by the formula). For odd `m` every prime of the conductor is `≡ ±1 (mod 8)`, so there is no 3-, 5-, 7- or 13-isogeny to walk: the only way off the crater is a 409- or a 1721-isogeny. Their division polynomials have degree 83,640 and 1,480,920; the torsion route found three kernels of each at each vertex through abscissae in `F_2^{41·204}` and `F_2^{41·215}` (eigenvalue orbit degrees 204 and 215), 54 Montgomery ladders of 8.3–8.8 kbit, 39 s each on a loaded core; a 409-kernel search took 197 s and a 1721-kernel search 978 s (three kernels each), so the transport to a conductor-703889 curve is 9–15 minutes against a 0.46 s solve. Every edge verified on points; `#E` re-counted on all 11 curves; the duals of the sampled edges, found by pushing a second torsion abscissa through them, carried the shifted instances back to the crater.
- **R1–R3:** all 11 curves within the control distribution (largest |z| 2.38 against 2.84); heterogeneity not detected (KW p = 0.21, ANOVA p = 0.076, variance ratio 1.70 on 11 curves); the four strata — the Koblitz crater (0.996), conductor 409 (1.009), 1721 (1.001) and 703889 (0.978) — within ±2.5 % of $\sqrt{(\pi n/2)}$ with |z| ≤ 0.8; both levels of both volcanoes likewise. Representation controls (three other polynomial bases) 0.98, 1.03, 0.96; A/A 0.97. The generic walk does not know it is on a Koblitz curve.
- **R4:** `plant-neg` 0.769 [0.705, 0.838] (z = −5.9); `plant-r4` 1.358 [1.221, 1.508] (z = +4.7). **`fold-frob` 0.155 [0.141, 0.170] against the expected 1/√41 = 0.156 (z = −48); `fold-frobneg` 0.113 [0.103, 0.124] against 1/√82 = 0.110 (z = −53).** Each folded step costs 41 squarings more (1.92 and 2.05 µs per step against 0.50 µs), and the fold still wins in wall time: 275 ms and 212 ms per solve against 459 ms. The fold needs `τ: (x, y) ↦ (x², y²)` to be an endomorphism, which it is only when the curve is defined over F_2; every other curve of the class has `b ∉ F_2`, no such map, and the same plain cost. The 409- and 1721-isogenies carry the instance away from the only curve where the √(2m) speed-up exists.
- **R5:** 3 curves × 64 generators × 25 seeds, 4,800 solves, all correct. Generator means spread as the null predicts on every curve (sd 94,978– 102,029 against 95,141–102,210 expected; variance ratios 1.00–1.04; KW p 0.34–0.72); no generator flagged (family 192, largest |z| 3.52 against 3.65); |ρ| ≤ 0.20 with every generator feature.
- **R6:** 3 sources (the crater, a conductor-409 and a conductor-1721 curve) × 11 targets × 25 seeds, 825 solves, all correct. η^2(target) = 0.016, η^2(source) = 0.0008; KW by target p = 0.36, by source p = 0.49; no cell flagged (largest |z| 3.11 against 3.17).

Across the four classes

Everything the protocol allowed the data to say about the curve, the stratum, the level, the route, the generator and the source came back *within the control distribution* every positive control came back *detected*, at the size the theory of the fold predicts. The spread of curve means was 1.1–1.7× the pooled single-solve prediction (1.18, 1.13, 1.11, 1.70 on 104, 45, 40, 11 curves); the largest is the smallest family, and no single curve carried it. The 32-adding walk itself ran 0.5–2.0 % above $\sqrt{(\pi n/2)}$ in every class, consistent with Teske's `1/√(1 − 1/32)`.

Limits

- One root per class; the measured curves are a stratified sample (eight per stratum where the stratum had that many within the walk radius, plus the curves reached through the largest degrees), not the whole class. Class sizes are exact; coverage is what the tables say.
- Operations are the unit. Wall times (ms, ns/op, transport, end-to-end) are from a shared 4-vCPU VM with other jobs running and are indicative; no isolation receipt (`isolated_bench`) was taken.
- Levels of the Koblitz class's sampled primes are assumed descending off the crater, exact there because `h(−7) = 1`; the `ℓ` = 193 prime of P0's conductor was not walked (`Φ_193` is beyond this session).
- Sampled kernels (torsion route) are three of `ℓ + 1`; the Koblitz class walk is a sample of a sample.
- This is rho against rho, one instrument. It is not an index-calculus result, has no §5a band, and says nothing about curves outside these four classes. A fold that needs the crater's structure is not an "isogeny speedup": it is the crater curve's own endomorphism, which the isogeny takes away.

Reproduce

```
cargo build --release --bin isoclass-rho
target/release/isoclass-rho selftest
# then the commands of runs/run-20261018T0857Z/COMMANDS.md, in order;
# python plot.py runs/run-20261018T0857Z/<class> and
# python make_report.py runs/run-20261018T0857Z draw and tabulate the records.
```

Replay: each class mode re-run with `--trials 5` reproduces the recorded iteration counts of the first five seeds (`<class>/replay/`, compared in the run README).